

AI-Driven Climate-Aware Decision Support and Risk Forecasting Systems for Sri Lanka

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Project Proposal Report
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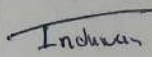
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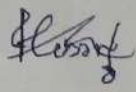
March 2026

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Floods are one of the most frequent and damaging natural disasters in Sri Lanka, affecting millions of people, destroying homes, and disrupting daily life. Despite significant research on flood monitoring and early warning systems, existing solutions often provide only general alerts and lack predictive analysis based on historical rainfall and river-level data. This limitation reduces their effectiveness in timely disaster preparedness and community response.

This research addresses the identified gap by developing an integrated flood prediction system that combines historical flood records, rainfall radar data, and river-level measurements to generate short-term hyper-local flood forecasts. The system leverages a CNN-LSTM machine learning model to capture both spatial and temporal patterns in rainfall and river flow, providing accurate predictions with lead times of up to six hours.

A React Native mobile application serves as the user interface, displaying interactive flood maps, dashboards, and alerts. Data storage, authentication, and notification services are managed through Firebase cloud infrastructure, ensuring scalability, security, and real-time performance. The system aims to make flood information accessible and actionable for local communities, disaster management authorities, and researchers.

Expected outcomes include improved early warning capabilities, reduced risk of property and life loss, and enhanced decision-making during flood events. Socially, the system can help communities respond proactively to imminent floods, while commercially, it can support government agencies, insurance companies, and NGOs involved in disaster management. By integrating historical data, machine learning and mobile visualization, this system represents a novel approach to predictive flood management in Sri Lanka.

Keywords: Flood Prediction, CNN-LSTM, Real-time Alerts, Disaster Management, Sri Lanka

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LIST OF ABBREVIATIONS

TABLE 1: List of Abbreviations and their full meaning

Abbreviation	Full Form
CNN-LSTM	Convolutional Neural Network – Long Short-Term Memory
GIS	Geographic Information System
ML	Machine Learning
SDG	Sustainable Development Goal

1. INTRODUCTION

1.1 Background and Context

Flood is one of the main natural disasters that happens in Sri Lanka. Each year it destroys lives, properties, and the day-to-day activities of people. Heavy rainfall during monsoon seasons causes rivers to overflow and leads to flooding in many parts of the country. Low-lying areas and regions located near river basins are especially vulnerable to flood disasters [1].

Floods affect thousands of people every year and cause serious social and economic damage. According to disaster management reports, severe flood events in Sri Lanka have affected hundreds of thousands of people and caused significant damage to infrastructure, agriculture, and housing. River basins such as the Kelani River basin frequently experience flooding due to heavy rainfall and increasing urban development in surrounding areas [2]. These disasters disrupt transportation systems, public services, and economic activities, making flood management an important national concern.

At the global level, floods are considered the most common natural disaster. According to global disaster reports, floods account for nearly 40% of all natural disasters worldwide and affect millions of people every year [3]. Climate change, extreme weather conditions, and rapid urbanization are increasing the frequency and intensity of flood events in many countries. As a result, improving flood monitoring and prediction systems has become an important research area worldwide.

Several researchers and government institutions have developed systems to monitor flood risks using rainfall measurements, hydrological modelling, and geographic information systems (GIS). Flood inundation models and rainfall-runoff models are commonly used to analyze flood patterns and identify flood-prone areas [4][5]. These methods help authorities understand flood behavior and support disaster management planning.

However, many existing systems mainly focus on rainfall monitoring or general flood warnings and may not provide clear visualization or predictive analysis that can help communities understand future flood risks. In many cases, people receive alerts only when heavy rainfall has already started, which limits their ability to prepare for potential disasters [6].

To address these limitations, this research proposes the development of a flood monitoring and prediction system that integrates historical rainfall data and river water level data. By analyzing historical patterns and relationships between rainfall intensity and river levels, the system can identify trends and estimate possible future flood situations in specific regions. The proposed system will include several components such as rainfall data analysis, river level monitoring, flood risk prediction, and visualization through an interactive map and dashboard. This approach can provide more accessible and location-specific information to help communities and authorities better prepare for potential flood events [7].

This research supports the United Nations Sustainable Development Goals, especially SDG 11 on Sustainable Cities and Communities, by aiming to reduce the impact of disasters on urban areas. It also contributes to SDG 13 on Climate Action by addressing climate-related hazards and helping communities prepare for natural disasters.

From a research perspective, this study belongs to the Centre of Excellence in Artificial Intelligence (CoEAI) research cluster, as it applies artificial intelligence techniques, environmental data analysis, predictive modelling, and spatial visualization methods to improve flood monitoring, prediction, and disaster risk management.

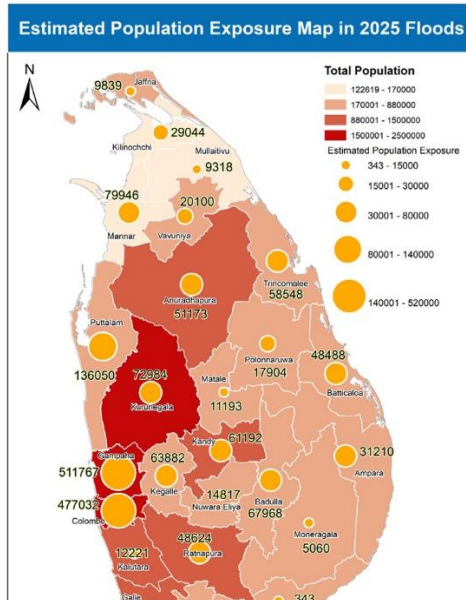


Figure 1: Estimated population exposure to 2025 floods in Sri Lanka.

1.2 Literature Review

Flood monitoring and prediction systems play an important role in disaster management and flood risk reduction. These systems generally integrate environmental data such as rainfall measurements, river water levels, and weather forecasts to monitor flood conditions and provide early warnings. Many modern flood management systems use hydrological models, data analysis techniques, and geographic information systems (GIS) to analyze flood risks and support decision-making. By combining environmental monitoring with analytical models, these systems help authorities identify flood-prone areas, understand flood patterns, and respond more effectively to potential flood events.

Floods are one of the most common natural disasters worldwide and have been widely studied using different scientific approaches. Researchers have focused on flood prediction, flood risk assessment, and flood monitoring systems to reduce the impact of flood disasters. This section reviews existing research approaches, current systems and services, and their limitations.

Many studies have used hydrological and hydraulic modelling techniques to understand flood behavior. For example, flood inundation modelling using HEC-RAS has been applied to simulate flood conditions and identify flood-prone areas in urban regions [1]. Such models are useful for analyzing river flow and predicting flood extent under different rainfall conditions. Similarly, flood risk assessments have been conducted in the Kelani River basin to evaluate the vulnerability of surrounding communities and infrastructure [2]. These studies show that river basin characteristics and rainfall intensity play a major role in flood occurrence.

Several researchers have also investigated social and environmental impacts of floods. Studies on urban flood resilience have examined how communities perceive and respond to flood disasters, highlighting the importance of improving disaster awareness and preparedness [6]. In addition, global disaster reports show that floods are responsible for a large proportion of natural disaster impacts worldwide, affecting millions of people each year [3]. This highlights the importance of developing better monitoring and prediction systems.

Another research approach focuses on hydrological modelling and prediction techniques. Rainfall-runoff models have been used to analyze the relationship between rainfall and river discharge levels [4]. These models help estimate how rainfall contributes to river flow and potential flooding. Dynamic flood simulation models have also been developed for river basins such as the Kelani River basin to understand flood patterns and flood propagation [5]. These methods provide useful insights into flood processes but often require complex modelling expertise and high computational resources.

Recent studies have also explored flood determinants and environmental factors. Research has shown that land use changes, urbanization, and environmental factors significantly influence flood risks in many regions [9]. Flood hazard mapping and floodplain studies have also been conducted to identify areas that are more vulnerable to flooding and to support disaster management planning [8].

In addition to academic research, several existing products and services have been developed to monitor flood risks. Government agencies such as disaster management authorities and meteorological departments provide rainfall forecasts and flood warnings to the public. These warning systems typically use rainfall monitoring data and weather forecasts to issue alerts through media channels and communication networks. Modern flood early warning systems have also been developed to support disaster risk reduction efforts [11]. However, many of these systems provide only general warnings and do not include detailed flood risk visualization or predictive analysis based on historical data and river level measurements.

Despite significant research and existing systems, several limitations remain. Many hydrological models are complex and require expert knowledge to operate, making them less accessible for general users. Existing warning systems also focus mainly on real-time alerts rather than predictive analysis based on historical trends. Additionally, many systems do not provide user-friendly visualization tools such as interactive maps or dashboards that allow communities to easily understand flood risks in specific locations.

Therefore, there is a need for an integrated flood monitoring and prediction system that combines historical rainfall data, river level measurements, and predictive analysis in a user-friendly platform. Such a system can improve accessibility to flood information and support better disaster preparedness for communities and authorities [7].

1.3 Research Gap

Flood disasters have been widely studied using hydrological modelling, flood risk assessment, and community resilience analysis. Previous studies have used flood inundation models, rainfall-runoff models, and floodplain mapping techniques to understand flood behavior and identify flood-prone areas [1][2][5]. These approaches provide valuable insights into flood patterns and support disaster management planning.

However, many of these studies mainly focus on hydrological simulations and risk analysis rather than providing accessible systems that combine multiple datasets to support practical flood prediction and decision-making. Hydrological models such as HEC-RAS and rainfall-runoff models often require complex configurations, specialized expertise, and large amounts of data, which limits their usability for general communities and non-technical users [1][4].

In addition, existing flood monitoring systems and early warning services provided by government agencies mainly focus on real-time rainfall monitoring and general flood alerts. These systems typically provide warnings after heavy rainfall has already occurred and do not fully utilize historical rainfall patterns and river level measurements to estimate possible future flood events. As a result, communities and decision-makers may not receive sufficient information to prepare for potential flood risks in advance.

Another limitation identified in existing research is the lack of user-friendly visualization platforms that integrate multiple data sources. Many flood modelling studies produce technical results or reports but do not provide interactive systems that allow users to easily visualize flood risks through maps, dashboards, or analytical tools. This limits the accessibility of flood-related information for the general public and local authorities.

Therefore, there is a clear research gap in developing an integrated flood monitoring and prediction system that combines historical rainfall data, river water level measurements, and predictive analysis in a single accessible platform. Such a system can analyze past flood patterns, identify relationships between rainfall and river levels, and estimate possible future flood risks for specific locations.

The proposed research addresses this gap by developing a data-driven flood prediction and visualization system that integrates historical environmental data with predictive analysis techniques. By providing interactive visualization and location-specific flood risk insights, the system aims to improve accessibility to flood information and support better disaster preparedness and decision-making.

2. OBJECTIVE

2.1 Main Objective

The main objective of this research is to develop a comprehensive flood monitoring and prediction system that uses historical rainfall data and river water level measurements to identify flood patterns and estimate potential future flood risks. The system will integrate data collection, predictive modelling, and interactive visualization, providing actionable flood risk information for communities, local authorities, and disaster management agencies. By combining historical trends with predictive analytics, the system aims to improve preparedness, reduce damage to lives and property, and support informed decision-making before and during flood events.

2.2 Specific Objectives

To achieve the main objective, the research is divided into several specific objectives, each addressing a key component of the proposed system.

The first specific objective is to collect and preprocess historical rainfall and river level data from trusted sources in Sri Lanka. This step is crucial to ensure data quality, completeness, and consistency, as the accuracy of flood predictions relies heavily on the quality of input data. Data preprocessing will include cleaning missing or inconsistent values, standardizing formats, and structuring datasets for analysis.

The second objective is to analyze historical rainfall and river level patterns to understand the relationships between rainfall intensity, river water levels, and flood occurrences. This analysis will identify thresholds or conditions that typically lead to flooding and highlight areas with the highest vulnerability. Understanding these relationships is essential for designing accurate predictive models.

The third objective is to develop a predictive model that estimates potential future flood risks based on historical data. This model will combine statistical analysis, ML, or other suitable predictive techniques to forecast flood likelihood in specific regions. The goal is to provide a forward-looking system rather than only monitoring real-time conditions, which will allow authorities and communities to prepare proactively.

The fourth objective is to design and implement a user-friendly visualization dashboard that presents flood risk information in an accessible way. The dashboard will display interactive maps, charts, and graphs to help users understand potential flood impacts and trends. By making complex data understandable, the platform will bridge the gap between technical flood modelling and practical decision-making.

The fifth objective is to evaluate the accuracy and effectiveness of the flood prediction system using historical flood events. This evaluation will measure how closely predicted flood risks match actual events, providing evidence of system reliability and identifying areas for improvement.

Finally, the research aims to develop a system that integrates all components into a single, functional platform. By combining data collection, analysis, prediction, and visualization, the system will provide an end-to-end solution for flood monitoring and risk assessment. This holistic approach ensures that the system is not only technically robust but also practical and usable for end users, including local communities, disaster management agencies, and policymakers.

3. METHODOLOGY

3.1 Research Approach

This research follows a Design Science approach, focusing on creating a practical and innovative flood prediction system for Sri Lanka. Design Science is suitable because the project involves developing an integrated artifact that collects, processes, and analyzes rainfall and river data to generate hyper-local flood forecasts, and then presents them via a user-friendly platform. The approach emphasizes iterative design, implementation, and evaluation, ensuring that the system is technically robust, scalable, and usable by disaster management authorities and local communities.

3.2 Technical Methods and Algorithms

The core predictive model is a CNN-LSTM network. The CNN component extracts spatial patterns from rainfall radar grids, capturing the geographic distribution and intensity of rainfall, while the LSTM component models temporal dependencies, learning how rainfall and river levels evolve over time.

Before training, the system preprocesses large datasets: raw radar rainfall data is cleaned, normalized, and temporally aligned with historical flood records from the Disaster Management Centre. Spatial features are extracted from radar grids, and time-series datasets are prepared for training the CNN-LSTM model. The trained model generates short-term flood forecasts for 0-6 hour lead times, providing actionable insights for flood-prone areas.

3.3 Tools and Platforms with Justification

The system is built using a combination of Python, TensorFlow, Firebase, React Native, and GIS tools, each selected based on its suitability for the project requirements.

Python and TensorFlow are used for data processing, feature extraction, and training the CNN-LSTM model. Python was chosen because of its extensive libraries for data analysis and ML, while TensorFlow was selected for its stability, GPU support, and ability to handle both spatial and temporal data efficiently. Alternative frameworks, such as PyTorch, were considered but were not used because TensorFlow integrates more easily with cloud deployment and production pipelines.

Firebase serves as the cloud backend, providing a real-time database, cloud storage, and user authentication. It was chosen over traditional SQL or NoSQL databases because it allows seamless real-time synchronization between the mobile application and the backend. This is essential for delivering timely flood alerts and updates. Other database solutions were not selected due to the additional setup required to achieve the same level of real-time performance and mobile integration.

React Native is used to develop the mobile application. Native development using Java, Kotlin, or Swift was not selected because it would significantly increase development time and maintenance effort. React Native also integrates directly with Firebase, simplifying the flow of data from the backend to the user interface.

Together, these tools provide a reliable, efficient, and user-friendly system, capable of processing large datasets, generating accurate predictions, and delivering timely information to end users.

3.4 Data Collection and Ethical Considerations

The system uses historical radar rainfall data, river water level records, and historical flood events from official Sri Lankan sources. Python pipelines automate data collection, cleaning, normalization, and temporal alignment to ensure high-quality datasets. Spatial features are extracted from radar grids to capture geographic patterns relevant to flood risk.

Ethical considerations include acknowledgment of data sources, responsible handling of sensitive location-specific information, and adherence to research best practices. Since Firebase stores user and forecast data, security rules are implemented to protect privacy and prevent unauthorized access.

3.5 Validation Strategy and Performance Metrics

The predictive system will be validated using historical flood events, comparing model predictions against real occurrences. Key performance metrics include:

Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) for predicted river levels. Precision, Recall, and F1-score for classifying flood and non-flood events.

Usability evaluation of the mobile app and dashboard, including map accuracy, clarity of alerts, and responsiveness.

This combination of technical and practical evaluation ensures that the system is accurate, reliable, and user-friendly, supporting both disaster management authorities and local communities in flood-prone regions.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

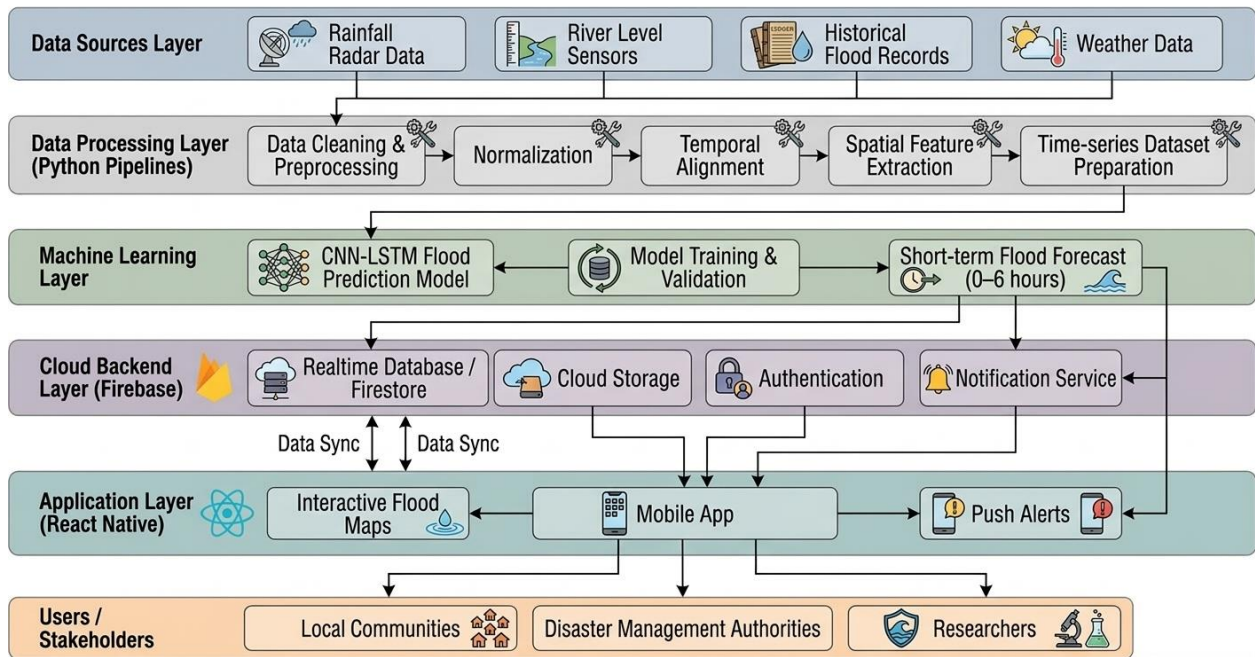


Figure 2: High-level System Architecture Diagram

5. USER REQUIREMENTS

5.1 Stakeholder Identification

Local Communities: - Residents in flood-prone areas need timely forecasts and alerts to protect lives and property. Without addressing their needs, the system would not fulfill its primary purpose of public safety.

Disaster Management Authorities: - Government agencies require accurate predictions to coordinate disaster response and issue warnings. Their input ensures the system is aligned with operational procedures.

Meteorological Department of Sri Lanka: - Provides rainfall radar and weather data, which is essential for accurate flood prediction. Integrating their datasets ensures reliability and scientific validity.

Urban Planners and Local Government Officials: - Use flood forecasts to design infrastructure and evacuation plans, preventing long-term damage and improving city resilience.

Researchers and Environmental Analysts: - Require historical and predictive data to study trends, analyze climate impacts, and support policy decisions.

5.2 Functional Requirements

1. Data Integration: - Collect and integrate historical rainfall, river level, and flood event data from trusted sources.

2. Data Preprocessing: - Clean, normalize, and temporally align raw datasets to prepare them for predictive modelling.

3. Flood Prediction: - Generate short-term, hyper-local flood forecasts (0-6 hours) using CNN-LSTM models.

4. Real-time Alerts: - Send flood warnings and notifications to mobile users based on predicted risks.

5. Interactive Visualization: - Display flood-prone areas, rainfall intensity, and risk levels on maps within the mobile application.

6. User Account Management: - Allow users to register, log in, and manage notification preferences.

7. Dashboard Analytics: - Provide historical trends, risk reports, and flood severity charts for authorized stakeholders.

8. System Administration: - Allow administrators to manage datasets, update models, and monitor system performance.

5.3 Non-Functional Requirements

- **Performance:** - The system must be able to process large volumes of rainfall radar data efficiently and generate predictions within a short time period. Flood predictions are time-sensitive, and delays in processing could reduce the usefulness of warnings. Therefore, the system should support near real-time processing so that users receive timely alerts before flooding occurs.
- **Security:** - The system must ensure secure handling of user data, prediction results, and system access. Authentication mechanisms should be implemented to prevent unauthorized access to the mobile application and administrative functions. Secure communication protocols such as HTTPS should be used when transferring data between the mobile application, cloud services, and prediction servers. This is important because disaster-related information must be reliable and protected from misuse or manipulation.
- **Usability:** -The mobile application must provide a simple and intuitive interface so that users can easily understand flood warnings and risk information. Flood prediction results should be displayed using clear maps, color indicators, and notifications. Since the system will be used by people with different technical backgrounds, the interface must minimize complexity and allow users to quickly interpret flood risk levels.
- **Reliability:** - The system must operate continuously and deliver consistent prediction results without frequent interruptions. During heavy rainfall events or emergency situations, system availability becomes critical. Reliable cloud infrastructure and proper system monitoring should be implemented to reduce downtime and ensure that warnings are delivered when they are most needed.
- **Scalability:** - The system should be able to handle increasing amounts of data and a growing number of users over time. Rainfall radar datasets can be very large, and the number of mobile users may increase during disaster events. Cloud platforms such as Firebase allow the system to scale resources dynamically, ensuring stable performance even when data volume and user demand increase.
- **Maintainability:** - The system should be designed using modular components so that updates and improvements can be implemented easily. For example, the ML model may need retraining when new rainfall data becomes available. A maintainable architecture ensures that datasets, prediction models, and application features can be updated without disrupting the entire system.

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

The proposed flood monitoring and prediction system targets three main user groups: government institutions, private organizations, and the general public living in flood-prone areas.

The primary customers are government agencies responsible for disaster management and environmental monitoring, such as the Disaster Management Centre, Department of Meteorology, and the Irrigation Department. These institutions require accurate flood monitoring tools to support emergency planning, early warnings, and disaster response.

The secondary customers include organizations that require flood risk information for planning and decision making. This includes insurance companies, infrastructure development organizations, urban planners, and construction companies operating in flood-prone regions.

In addition, the general public represents an important group of end users. People living near rivers or in low-lying areas are directly affected by floods and require timely information about potential flood risks. Through the mobile application, citizens can access real-time flood alerts, rainfall information, and flood risk maps, which can help them take precautionary actions such as protecting property or preparing for evacuation.

6.2 Value Proposition

The system provides a data-driven flood monitoring and prediction platform that integrates rainfall data, river water level measurements, and historical flood records. Unlike traditional monitoring approaches that rely mainly on rainfall observations, the proposed system combines multiple environmental data sources with predictive modelling techniques to estimate possible future flood conditions.

For government institutions, the system supports better disaster preparedness, faster decision-making, and improved flood risk assessment. For organizations, the platform provides valuable environmental insights for infrastructure planning and risk management. For the general public, the system offers easy access to early warnings and flood risk visualization, which can help communities respond more effectively to flood threats.

6.3 Revenue Model

The system will primarily generate revenue through institutional licensing and deployment agreements with government agencies and organizations that require advanced flood monitoring capabilities.

- Government system licensing: - Institutions can purchase the system for official disaster monitoring and planning activities.
- Enterprise licensing: - Insurance companies, research organizations, and urban planning agencies may license the system for flood risk analysis.
- Public access services: - While basic flood alerts will remain free for citizens, additional features such as detailed flood analytics or historical flood reports may be offered through a small service fee.

This model ensures that critical flood warning information remains accessible to the public while maintaining financial sustainability through institutional partnerships.

6.4 Cost Estimation

The main operational costs include cloud infrastructure, database management, and system maintenance.

TABLE 2: Cost Estimation Table

Cost Component	Estimated Monthly Cost (LKR)	Annual Cost (LKR)
Cloud computing for data processing	6,000	72,000
Database and storage services	3,000	36,000
Data storage and backup	2,000	24,000
Security and encryption services	1,000	12,000
System maintenance and monitoring	30,000	360,000

Total Estimated Annual Operational Cost: LKR 504,000

6.5 Pricing Strategy

The pricing model focuses mainly on institutional licensing, while keeping essential flood alerts available for the public.

TABLE 3: Pricing Strategy Table

Service Type	Price (LKR)	Description
Government institutional license	1,500,000	Full system deployment and training
Enterprise organization license	1,200,000	Flood monitoring and analytics access
Advanced flood analysis reports	150,000 per report	Custom risk assessment for projects
Public mobile application	Free	Basic flood alerts and map visualization

This strategy ensures wide public access to flood warnings while generating revenue from organizations that require advanced analytical capabilities.

6.6 Competitive Advantage

The proposed system offers several advantages compared with many existing flood monitoring systems.

- Integration of rainfall data, river water levels, and historical flood information.
- Predictive modelling capabilities to estimate potential flood situations.
- Interactive map visualization and mobile application access.
- Cloud-based infrastructure that allows scalable deployment across multiple regions.

These features make the system useful for both professional disaster management authorities and ordinary citizens living in flood-prone areas.

6.7 Intellectual Property Considerations

The intellectual property associated with the proposed system includes the predictive flood modelling algorithms, data processing pipelines, and system architecture used for flood monitoring and visualization.

The software implementation and predictive models can be protected under software copyright and licensing agreements. Additionally, institutional deployments will operate under formal licensing contracts, ensuring that the system is used in accordance with the intellectual property rights of the developers. External datasets used within the system will comply with the data access policies of the relevant government agencies and research organizations.

7. BUDGET AND JUSTIFICATION

This section outlines the estimated budget required for the successful development and deployment of the flood prediction system. As a student-level research project, the budget is kept minimal by utilizing free and open-source tools, existing personal hardware, and free-tier cloud services wherever possible.

7.1 Hardware Costs

The researcher will use an existing personal laptop, eliminating the need for new hardware.

7.2 Software Licenses

TABLE 4: Software Licenses Cost Table

Software	Purpose	License Type	Cost (LKR)
Python 3.x	Data processing & ML model	Free / Open Source	0
TensorFlow	CNN-LSTM model training	Free / Open Source	0
React Native	Mobile app development	Free / Open Source	0
VS Code	Code editor	Free	0
Google Colab Pro	GPU-accelerated model training	Paid (monthly)	25,000
TOTAL			25,000

All core development tools Python, TensorFlow, and React Native are free and open-source, making them ideal for a student project. Google Colab Pro is a low-cost alternative to purchasing a high-end GPU, providing sufficient computing power for training the CNN-LSTM model within a limited budget.

7.3 Cloud Services

TABLE 5: Cloud Services Cost Table

Service	Usage	Plan	Cost (LKR)
Firebase Spark Plan	Database, auth, notifications	Free tier	0
Firebase Blaze Plan	Cloud Functions (if needed)	Pay-as-you-go	5,000
Google Maps API	Flood map visualization	Free tier (limited)	0
TOTAL			5,000

Firebase's free Spark Plan covers the core needs of this project, including real-time database, authentication, and push notifications. The Blaze Plan is budgeted as a contingency in case Cloud Functions are required for automated model inference. Google Maps API free tier is sufficient for displaying flood maps within the expected usage limits of a student prototype.

7.4 Human Effort Cost

TABLE 6: Human Effort Cost Table

Role	Task	Cost (LKR)
Printing & Binding	Reports and Research Documents	2,000
TOTAL		2,000

Printing and binding costs are estimated for reports and research documents to the Department of Information Technology, SLIIT.

7.5 Budget Summary

TABLE 7: Budget Summary Table

Category	Total Cost (LKR)
7.1 Hardware Costs	-
7.2 Software Licenses	25,000
7.3 Cloud Services	5,000
7.4 Human Effort Cost	2,000
GRAND TOTAL	LKR 29,000

The total estimated budget for this research project is LKR 29,000. This is a cost-effective budget achieved by prioritizing free and open-source software, utilizing personal hardware, and taking advantage of free-tier cloud services. The minimal financial investment reflects the student-level scope of this project while still maintaining the technical requirements necessary to build a functional flood prediction and visualization system.

8. WORK BREAKDOWN STRUCTURE (WBS)

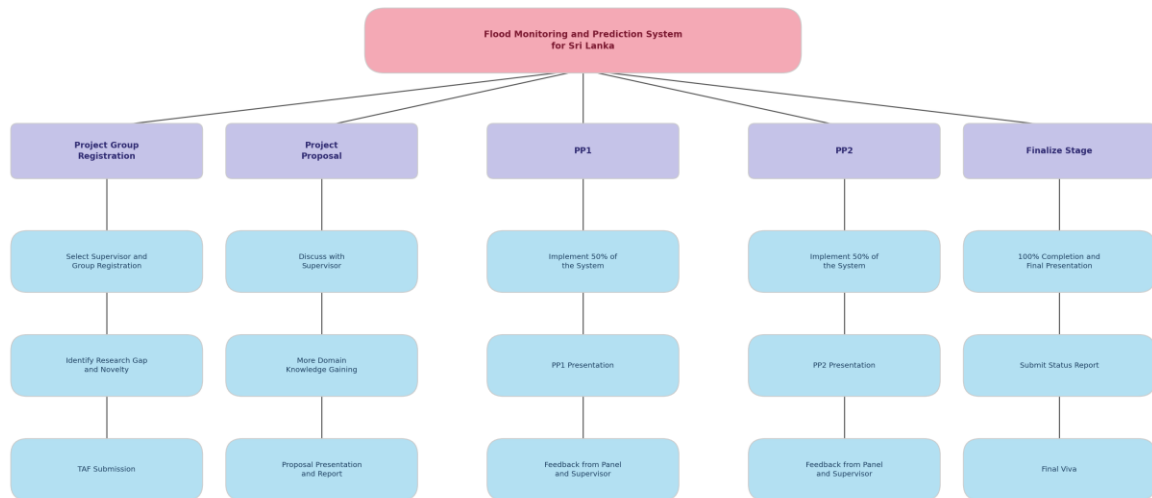


Figure 3: Work Breakdown Structure Diagram

9. GANTT CHART

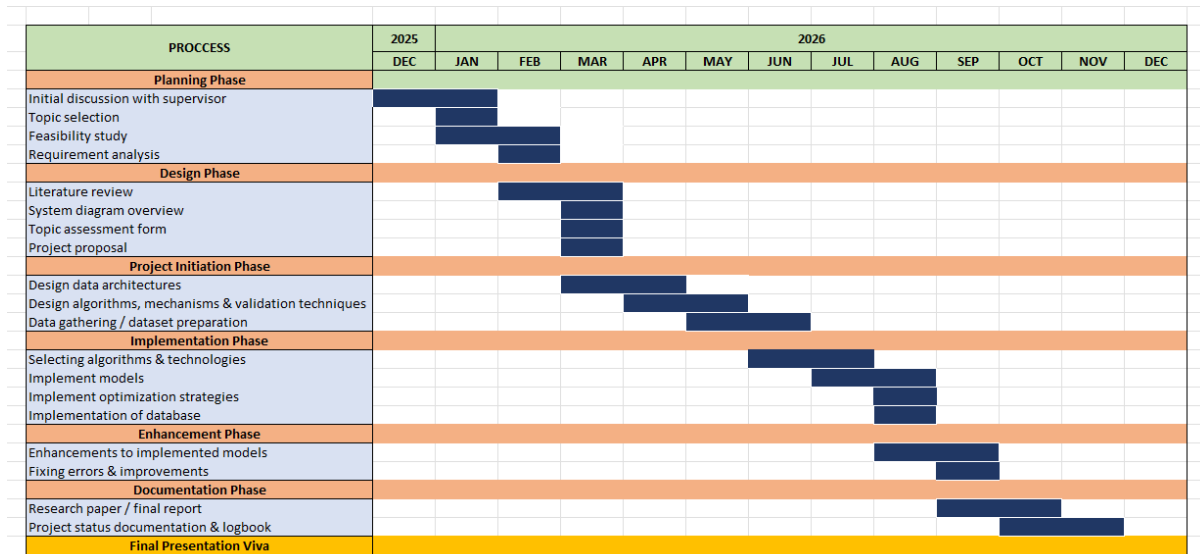


Figure 4: Gantt Chart

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AI-Driven Climate-Aware Decision Support and Risk Forecasting Systems for Sri Lanka

R26-IT-046

Project Proposal Report

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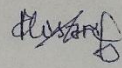
DECLARATION

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Dengue continues to be one of the most significant climate-sensitive public health threats in Sri Lanka, causing recurring outbreaks that place a heavy burden on healthcare facilities, public health authorities, and communities. The transmission dynamics of dengue are strongly associated with climatic factors such as rainfall, temperature, and humidity, which influence mosquito breeding, vector survival, and viral replication. Although Sri Lanka has established surveillance and reporting mechanisms for dengue through the Epidemiology Unit and the National Dengue Control Unit, these systems mainly support retrospective monitoring and routine reporting rather than forward-looking predictive decision support.

This study proposes an AI-driven climate-informed dengue risk forecasting and early warning system for Sri Lanka that integrates official dengue case records, climate variables, and geospatial context to generate short-term predictive insights. The proposed system is designed to operate primarily at district or RDHS level, where official surveillance data are more consistently available and practically useful for intervention planning. The research will involve data collection, preprocessing, temporal and spatial alignment, feature engineering, predictive modeling, hotspot identification, and risk communication through a mobile-accessible application. The system will transform historical disease and environmental data into weekly dengue risk forecasts, hotspot-oriented warnings, map-based visualizations, and concise public health insight summaries.

The technical approach will compare baseline statistical forecasting methods, machine learning models, and an LSTM-based sequential deep learning model in order to identify the most effective and operationally realistic forecasting approach for Sri Lanka. Forecast outputs will be delivered through a mobile application built using React Native, while Firebase services will support user-facing storage, synchronization, and alert dissemination. The final artifact is intended not as a replacement for official surveillance systems, but as a practical decision-support layer that enhances climate-aware public health planning and early intervention capacity. The expected contribution of the study is a Sri Lanka-specific, feasible, and scalable dengue early warning component that improves preparedness, prioritization, and public risk awareness.

Keywords— *Dengue Forecasting, Climate-Informed Disease Modeling, Machine Learning, Early Warning Systems, Geospatial Risk Mapping, Public Health Decision Support.*

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SSS

LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
ML	Machine Learning
LSTM	Long Short-Term Memory
ARIMA	Autoregressive Integrated Moving Average
RMSE	Root Mean Square Error
MAE	Mean Absolute Error
AUC	Area Under the Curve

1. INTRODUCTION

1.1. Background and Context

Climate change has intensified the frequency and severity of environmental hazards worldwide, particularly in tropical developing countries. Sri Lanka has become increasingly vulnerable to climate-induced risks such as extreme rainfall events, flooding, and vector-borne diseases. Among these, dengue fever has emerged as one of the most significant public health concerns. According to the Sri Lankan Epidemiology Unit, tens of thousands of dengue cases are reported annually, with major outbreaks recorded in recent years due to favorable climatic conditions that support mosquito breeding and virus transmission. Factors such as temperature fluctuations, prolonged rainfall, and high humidity significantly influence the lifecycle of *Aedes aegypti*, the primary dengue vector, making dengue transmission highly sensitive to climate variability [1], [2]. Consequently, climate-driven disease dynamics present a growing challenge for public health management and early warning systems.

The impacts of dengue outbreaks extend beyond public health. Frequent outbreaks place substantial pressure on healthcare infrastructure, disrupt economic productivity, and increase the vulnerability of communities, particularly in densely populated urban areas. Public health authorities, municipal planners, and environmental agencies require timely and localized insights to effectively allocate resources, implement vector control measures, and issue preventive warnings. However, existing monitoring systems in Sri Lanka remain largely reactive and fragmented. Traditional surveillance approaches rely heavily on retrospective case reporting and limited spatial analysis, which often delays outbreak detection and reduces the effectiveness of intervention strategies. Furthermore, many existing systems lack the capability to integrate heterogeneous datasets such as climate observations, epidemiological records, and environmental indicators in real time.

Recent advances in AI and data-driven forecasting present new opportunities to address these limitations. ML and deep learning techniques have demonstrated strong potential in modeling complex spatiotemporal relationships between climate variables and disease transmission patterns. However, most current dengue prediction models are either geographically generalized or limited by low spatial resolution and insufficient integration of local environmental data. As a result, there remains a significant need for scalable and context-specific forecasting systems tailored to the climatic and epidemiological characteristics of Sri Lanka.

To address this gap, this research proposes the development of an AI-driven dengue risk forecasting component within a broader climate-aware decision support platform for Sri Lanka. The system integrates meteorological datasets and historical dengue case records to model climate-disease relationships using advanced deep learning techniques, including LSTM networks and Transformer-based architectures. Through spatiotemporal data processing and feature engineering, the system aims to generate high-resolution weekly dengue risk predictions and risk maps to support proactive public health interventions.

The proposed research aligns with several United Nations Sustainable Development Goals (SDGs), particularly SDG 3 (Good Health and Well-Being), SDG 11 (Sustainable Cities and Communities), and SDG 13 (Climate Action). Additionally, the study contributes to the research agenda of the Centre of Excellence in AI (CoEAI) by advancing AI-driven solutions for disaster management, public health informatics, and climate-aware decision support systems tailored to national development needs.

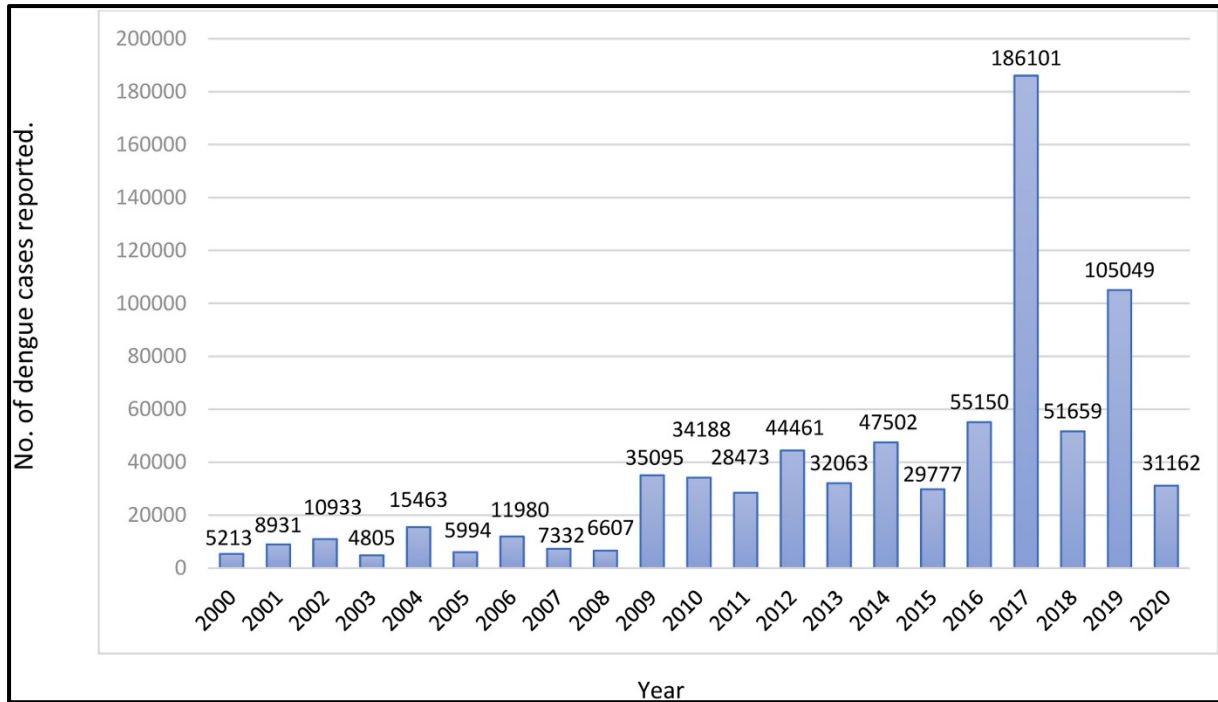


Figure 1: Reported Dengue Cases Chart (2000 - 2020)

1.2.Literature Review

Climate-sensitive diseases such as dengue have become a major global public health concern due to their strong relationship with environmental and climatic conditions. Dengue transmission is highly influenced by factors such as rainfall, temperature, and humidity, which affect the breeding patterns and lifecycle of *Aedes aegypti* mosquitoes. Studies indicate that climate variability significantly contributes to the expansion of dengue transmission zones, particularly in tropical regions such as South and Southeast Asia [3]. As a result, researchers have increasingly explored data-driven forecasting approaches that integrate climate data and epidemiological records to predict dengue outbreaks and support early warning systems.

Recent advances in AI and ML have enabled the development of predictive systems capable of modeling complex relationships between environmental factors and disease incidence. These systems typically follow a structured pipeline involving data acquisition, preprocessing, feature engineering, predictive modeling, and visualization of risk indicators. By analyzing historical climate and disease data, such systems aim to detect patterns that precede dengue outbreaks and provide timely insights for public health authorities [4]. However, despite significant advancements in predictive modeling techniques, many existing forecasting systems still face challenges related to spatial resolution, real-time data integration, and operational scalability.

Early research on dengue forecasting relied primarily on statistical time-series models such as ARIMA and Seasonal ARIMA. These models analyze historical case data to identify temporal trends and seasonal patterns in disease outbreaks. Although statistical models provide useful baseline predictions, they often struggle to capture nonlinear relationships between climate variables and disease transmission dynamics [5]. As dengue transmission depends on complex interactions among environmental, biological, and social factors, purely statistical approaches may produce limited forecasting accuracy when applied to large heterogeneous datasets.

To overcome these limitations, researchers have increasingly applied ML algorithms such as Random Forest, Support Vector Machines, and Gradient Boosting models. These algorithms can model nonlinear relationships and interactions between multiple climatic and environmental variables. Studies have shown that ML-based dengue prediction models can achieve improved forecasting accuracy compared with traditional statistical approaches when trained on large datasets containing meteorological and epidemiological variables [6], [7].

More recently, deep learning techniques have been introduced to further improve dengue forecasting performance. Recurrent Neural Networks (RNNs) and LSTM networks are particularly effective in modeling temporal dependencies within sequential data. Several studies have demonstrated that LSTM-based models can successfully predict dengue incidence by incorporating lagged climate variables such as rainfall, temperature, and humidity [8]. These models are capable of capturing long-term temporal patterns in climate and disease dynamics, leading to more accurate outbreak predictions compared with conventional ML approaches.

In addition to predictive modeling studies, several operational dengue monitoring platforms have been developed worldwide. Public health surveillance systems collect epidemiological data and generate reports for disease monitoring and outbreak response. Some modern platforms also incorporate geospatial analytics and satellite-derived environmental data to identify dengue risk hotspots and support vector control planning [9]. These systems provide valuable information for health authorities; however, many of them rely heavily on retrospective case reporting rather than predictive analytics.

Despite these technological advancements, several limitations remain in existing dengue forecasting systems. Many models are developed for specific geographic regions and may not generalize well to different climatic environments. In addition, some forecasting models rely on limited datasets or coarse spatial resolution, which reduces their effectiveness for localized outbreak detection. Data fragmentation between climate agencies and public health institutions also presents a major challenge for integrating heterogeneous datasets into a unified forecasting platform [10], [11]. Furthermore, many existing systems lack the capability to capture complex spatiotemporal interactions between environmental variables and disease transmission patterns across multiple regions simultaneously [12].

These limitations highlight the need for integrated AI-driven forecasting systems capable of combining climate data, epidemiological records, and advanced deep learning techniques. Developing such systems for climate-vulnerable countries like Sri Lanka can significantly improve early warning capabilities and enable proactive public health interventions to reduce the impact of dengue outbreaks.

1.3. Research Gap

Despite the growing body of research on dengue prediction and climate-based disease modeling, several critical gaps remain in the current literature and existing forecasting systems. While many studies have demonstrated the usefulness of statistical, ML, and deep learning models for predicting dengue outbreaks, most of these approaches are developed using datasets from specific geographic regions or controlled experimental environments. Consequently, their applicability to different climatic and socio-environmental contexts, such as Sri Lanka, remains limited. Climate-disease relationships vary significantly across regions due to differences in rainfall patterns, urban density, environmental conditions, and mosquito ecology. Therefore, models developed for other countries may not accurately capture the local transmission dynamics of dengue in Sri Lanka [6], [8].

Another major limitation in current approaches is the lack of integrated data platforms capable of combining heterogeneous datasets in a unified forecasting framework. Many existing dengue prediction studies rely on a limited number of variables, typically focusing only on historical dengue case counts and a small set of meteorological indicators such as rainfall or temperature. However, dengue transmission is influenced by complex interactions among multiple environmental and temporal factors. Without comprehensive data integration and spatiotemporal alignment, forecasting models may fail to fully capture these relationships, leading to reduced predictive performance and limited operational usability [4], [10].

Furthermore, many existing forecasting models operate at relatively coarse spatial and temporal resolutions. Several studies generate predictions at national or regional levels, which limits their usefulness for localized decision-making and targeted vector control interventions. Public health authorities require high-resolution predictions that can identify potential outbreak hotspots at district or sub-district levels. However, achieving such fine-grained predictions requires advanced spatiotemporal modeling techniques capable of processing large-scale climate and epidemiological datasets [7], [12].

Another research gap lies in the limited use of advanced deep learning architectures that can capture long-term temporal dependencies in climate–disease relationships. While traditional statistical models and conventional ML algorithms provide useful baseline predictions, they often struggle to model complex nonlinear relationships and delayed effects between climatic variables and dengue incidence. Deep learning models such as LSTM networks and Transformer-based architectures offer improved capabilities for modeling sequential data and long-range dependencies, yet their application in dengue forecasting remains relatively limited, particularly in the context of developing countries with fragmented data infrastructure [8], [11].

In addition, many existing dengue monitoring systems are designed primarily for retrospective analysis rather than proactive decision support. Public health surveillance platforms often rely on manual reporting of dengue cases, which can introduce delays in data availability and reduce the effectiveness of early warning mechanisms. As a result, health authorities may not receive timely risk predictions necessary to implement preventive measures such as vector control campaigns, resource allocation, or public awareness initiatives [9].

To address these challenges, this research proposes an AI-driven dengue risk forecasting system that integrates climate data and epidemiological records within a scalable spatiotemporal modeling framework. By combining advanced deep learning techniques with high-resolution climate datasets, the proposed system aims to generate weekly dengue risk predictions and spatial risk maps tailored specifically to the Sri Lankan context. This approach introduces novelty by leveraging both LSTM and Transformer-based architectures to capture complex climate–disease interactions while enabling localized outbreak forecasting. The system is designed to support proactive public health decision-making by providing actionable insights to health authorities, thereby contributing to improved disease prevention and climate-resilient healthcare planning.

Overall, the proposed research addresses critical limitations in existing dengue prediction systems by introducing an integrated, climate-aware, AI-driven forecasting framework that improves spatial resolution, data integration, and predictive accuracy. This contribution is expected to enhance early warning capabilities and support data-driven public health interventions in Sri Lanka.

2. OBJECTIVE

2.1 Main Objective

To develop an AI-driven dengue risk forecasting system that integrates climate variables and epidemiological data to predict potential dengue outbreaks in Sri Lanka and generate high-resolution weekly risk maps to support proactive public health decision-making.

2.2 Specific Objectives

- a) To collect and integrate climate and epidemiological datasets including rainfall, temperature, humidity, and historical dengue case records from national and global data repositories relevant to Sri Lanka.
- b) To preprocess and align heterogeneous datasets by performing missing value handling, temporal smoothing, spatial interpolation, and normalization to create a consistent spatiotemporal dataset suitable for predictive modeling.
- c) To engineer climate-based predictive features such as lagged rainfall patterns, temperature anomalies, and humidity trends that influence mosquito breeding and dengue transmission dynamics.
- d) To develop and train spatiotemporal deep learning models, including LSTM and Transformer-based architectures, using TensorFlow to capture temporal dependencies between climatic conditions and dengue incidence.
- e) To evaluate the performance of predictive models using quantitative metrics such as RMSE, MAE, and AUC, while comparing results with baseline models such as Random Forest and ARIMA.
- f) To generate high-resolution weekly dengue risk maps that identify potential outbreak hotspots across Sri Lanka and provide actionable insights for public health authorities to support early intervention and resource allocation.

3. METHODOLOGY

3.1 Research Approach

This research adopts a Design Science Research (DSR) approach, which focuses on designing and developing an innovative technological artifact to address a real-world problem. The artifact in this study is an AI-driven dengue risk forecasting system that integrates climate data and epidemiological records to generate predictive insights for Sri Lanka. The Design Science approach is appropriate because the research not only investigates theoretical relationships between climate variables and dengue transmission but also produces a practical decision-support tool that can assist public health authorities in early outbreak detection.

The methodology follows an iterative process consisting of problem identification, system design, model development, evaluation, and refinement. First, historical climate and dengue datasets are collected and analyzed to identify patterns associated with disease transmission. Next, predictive models are designed and trained using spatiotemporal deep learning techniques. Finally, the system is evaluated using established performance metrics to assess prediction accuracy and reliability. This iterative development process ensures that the resulting system is both scientifically valid and practically applicable for public health decision-making.

3.2 Technical Methods and Algorithms

The predictive component of the system is based on spatiotemporal deep learning models, specifically LSTM (LSTM) networks and Transformer-based architectures. These models are selected due to their ability to capture complex temporal dependencies between environmental conditions and dengue incidence.

The system processes multiple datasets, including rainfall, temperature, humidity, and historical dengue case records. Data preprocessing is performed to ensure data quality and consistency. This includes missing value imputation, temporal smoothing, normalization, and spatial interpolation to align meteorological and epidemiological data into consistent time intervals and spatial grids.

Feature engineering is then applied to derive climate-related indicators that influence dengue transmission. These include lagged rainfall values, temperature anomalies, humidity trends, and seasonal climate patterns. Such features help the model capture delayed environmental effects that influence mosquito breeding and virus transmission cycles.

The LSTM model is used to capture temporal relationships in climate and dengue datasets by learning sequential patterns over time. In addition, Transformer-based architectures are explored to improve long-range temporal forecasting and handle large time-series datasets efficiently. Baseline models such as Random Forest and ARIMA are also implemented for comparative evaluation. The trained models generate weekly dengue risk predictions, which are then visualized as spatial risk maps to identify high-risk regions across Sri Lanka.

3.3 Tools and Platforms with Justification

The system is implemented using a combination of Python, TensorFlow, Firebase, React Native, and NoSQL databases, selected based on their ability to support large-scale data processing, ML model deployment, and real-time information delivery.

Python is used for data preprocessing, feature engineering, and model development due to its extensive ecosystem of scientific computing and ML libraries such as Pandas, NumPy, and

Scikit-learn. Python is widely adopted in data science and provides efficient tools for handling large environmental datasets.

TensorFlow is used to develop and train deep learning models including LSTM and Transformer architectures. TensorFlow was selected because it offers strong support for large-scale ML workflows, GPU acceleration, and production-ready deployment capabilities. Although frameworks such as PyTorch provide similar functionality, TensorFlow was chosen due to its stability and compatibility with scalable deployment pipelines.

For backend infrastructure, Firebase is used as the cloud-based platform for storing and synchronizing prediction outputs and risk alerts. Firebase provides real-time database capabilities, authentication services, and scalable cloud storage. It was selected over traditional relational database systems because it allows seamless real-time updates and easier integration with mobile applications.

The system uses a NoSQL database structure to store heterogeneous datasets such as prediction results, spatial risk indicators, and user interaction data. NoSQL databases are more suitable than traditional SQL databases in this context because they provide flexible schemas and efficient handling of semi-structured data generated by AI models.

A React Native mobile application is developed to deliver dengue risk alerts and visual risk maps to users. React Native was chosen because it enables cross-platform mobile application development using a single codebase. Compared to native development using Java, Kotlin, or Swift, React Native significantly reduces development time while maintaining acceptable performance. Additionally, it integrates efficiently with Firebase services, enabling real-time data updates and notifications.

Together, these technologies create a scalable platform capable of processing climate data, generating predictive insights, and delivering timely information to public health authorities and end users.

3.4 Data Collection and Ethical Considerations

The research uses climate and epidemiological datasets obtained from official and publicly available sources. Meteorological data such as rainfall, temperature, and humidity are collected from national meteorological agencies and global climate data repositories. Historical dengue case records are obtained from the Sri Lankan Ministry of Health and the Epidemiology Unit, which maintain national disease surveillance databases.

Automated data pipelines developed in Python are used to perform data cleaning, normalization, and temporal alignment to ensure the datasets are suitable for predictive modeling. Spatial aggregation techniques are also applied to align climate data with administrative regions where dengue cases are reported.

Ethical considerations are addressed throughout the research process. Data sources are properly acknowledged, and publicly available datasets are used whenever possible to ensure transparency. Sensitive health-related data is handled responsibly and aggregated at appropriate spatial levels to prevent identification of individuals or households. In addition, Firebase security rules are implemented to protect stored prediction results and prevent unauthorized data access within the system.

3.5 Validation Strategy and Performance Metrics

The predictive models are validated using historical dengue outbreak data to evaluate their forecasting accuracy. The dataset is divided into training, validation, and testing subsets, allowing the models to learn patterns from historical records while ensuring unbiased performance evaluation.

Model performance is assessed using several quantitative metrics. RMSE and MAE are used to measure the accuracy of predicted dengue case counts compared with observed values. Additionally, Area Under the Receiver Operating Characteristic Curve (AUC-ROC) is used to evaluate the model's ability to correctly classify high-risk and low-risk regions.

Comparative evaluation is conducted by testing the proposed deep learning models against baseline forecasting models such as Random Forest and ARIMA. This comparison helps determine whether the proposed AI-driven approach provides measurable improvements in predictive performance.

In addition to technical evaluation, the usability of the system interface and mobile application will also be assessed. The clarity of dengue risk maps, responsiveness of alerts, and accessibility of the mobile platform are considered to ensure the system provides meaningful and actionable insights for public health authorities and other stakeholders.

This comprehensive validation strategy ensures that the proposed dengue forecasting system is accurate, reliable, and practically applicable for early outbreak detection and climate-informed public health decision-making in Sri Lanka.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview of the Proposed Solution

The proposed system is an AI-driven dengue risk forecasting component within the broader AI-Driven Climate-Aware Decision Support and Risk Forecasting Platform for Sri Lanka. The architecture is designed to integrate heterogeneous datasets, perform large-scale climate–health data processing, and generate predictive insights for public health decision-making.

The system consists of five main modules: Data Acquisition Layer, Data Processing Layer, Feature Engineering Layer, Predictive Modeling Layer, and Decision Support & Visualization Layer. These components work together to transform raw climate and epidemiological data into actionable dengue risk predictions and spatial risk maps. The architecture enables automated data processing, scalable ML model training, and real-time forecasting capabilities.

4.1. System Components

The high-level system architecture includes the following major components:

1. Data Acquisition Layer

This layer collects raw datasets from multiple sources. Climate data such as rainfall, temperature, and humidity are obtained from national meteorological databases and global climate repositories. Epidemiological data, including historical dengue case records, are obtained from the Sri Lankan Ministry of Health and the Epidemiology Unit. The collected data may also include geospatial boundary data used for spatial mapping and risk visualization.

2. Data Preprocessing and Integration Layer

Raw datasets often contain missing values, inconsistent formats, and different temporal or spatial resolutions. This layer performs data cleaning operations including missing value imputation, normalization, temporal smoothing, and spatial interpolation. After cleaning, spatiotemporal aggregation techniques are used to align meteorological and epidemiological data into consistent time intervals (e.g., weekly) and geographic grids.

3. Feature Engineering Module

In this stage, climate-related features that influence dengue transmission are derived from the processed datasets. Examples include lagged rainfall variables, temperature anomalies, humidity trends, and moving averages of climate indicators. These engineered features help capture environmental patterns associated with mosquito breeding and virus transmission dynamics.

4. Predictive Modeling Module

The predictive modeling component implements ML and deep learning models to forecast dengue risk. LSTM (LSTM) networks are used to capture temporal dependencies in climate and disease patterns, while Transformer-based architectures are explored to improve long-range temporal forecasting capabilities. Baseline models such as Random Forest and ARIMA are also implemented for performance comparison and validation.

5. Decision Support and Visualization Layer

The final module converts model predictions into meaningful insights for stakeholders. The system generates weekly dengue risk maps, hotspot identification reports, and predictive indicators that can assist health authorities in implementing proactive public health interventions. Visualization dashboards allow decision-makers to interpret forecasting outputs easily and support data-driven resource allocation.

4.2.High-Level System Diagram

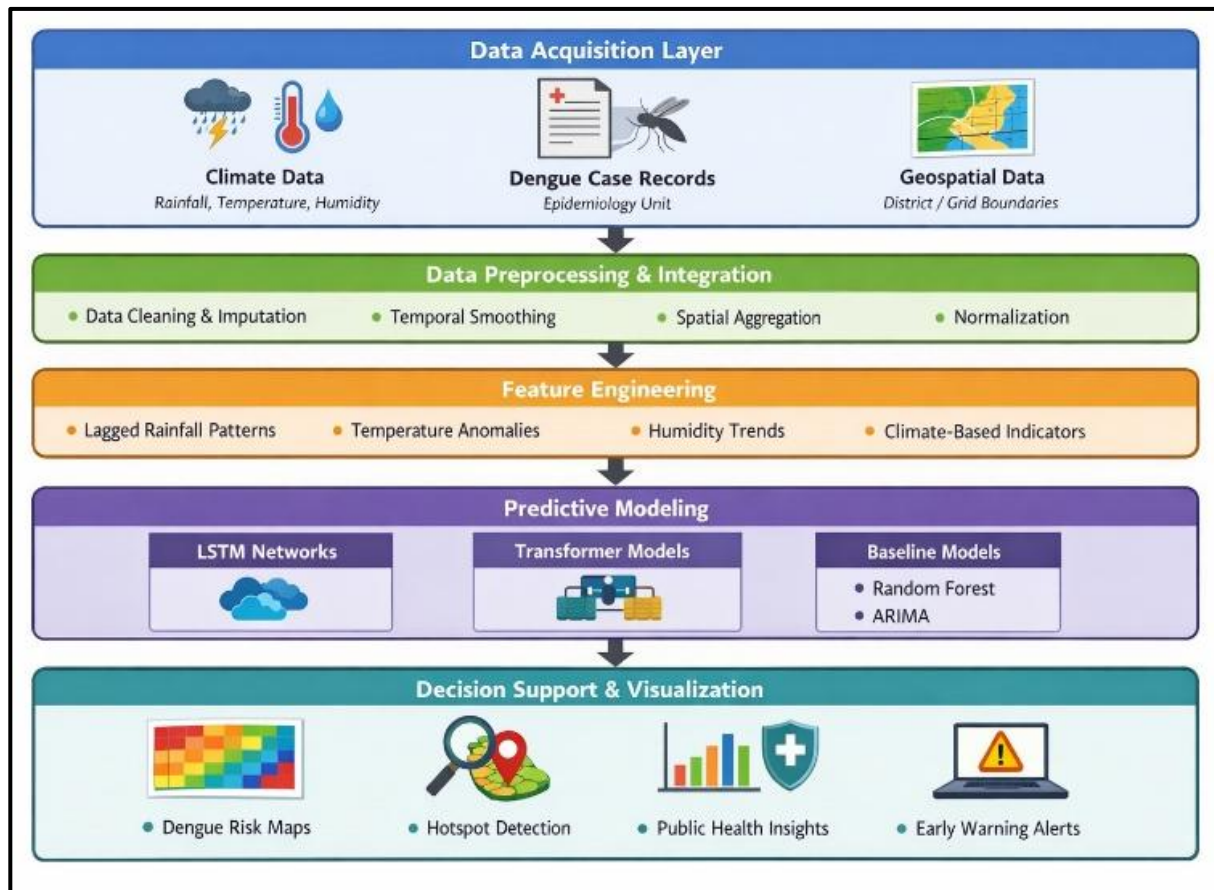


Figure 2 : High-Level System Architecture Diagram

4.3.Component Interactions

Each system component interacts sequentially to transform raw data into predictive insights. The Data Acquisition Layer continuously collects environmental and epidemiological datasets. These datasets are passed to the Data Processing Layer, where they are cleaned and standardized. The processed datasets are then forwarded to the Feature Engineering Module, which extracts climate-related predictors relevant to dengue transmission.

The engineered features serve as inputs to the Predictive Modeling Module, where deep learning and ML algorithms analyze temporal patterns and generate dengue risk forecasts. Finally, the predicted outputs are transmitted to the Decision Support Layer, which presents the results through visualization dashboards and spatial risk maps for public health authorities.

4.4.Data Flow Explanation

The data flow within the system follows a structured pipeline. First, raw climate and epidemiological datasets are collected and stored in the data repository. These datasets undergo preprocessing operations to remove inconsistencies and align spatial and temporal structures. After preprocessing, feature extraction algorithms generate climate-based indicators that serve as inputs to the predictive models.

The trained deep learning models process these features to forecast dengue risk levels for different geographic regions. The predicted results are then converted into visual outputs such as heat maps, dashboards, and risk reports. These outputs support early warning systems and assist health authorities in implementing proactive dengue prevention strategies.

4.5.Scalability and Security Considerations

The system architecture is designed with scalability in mind to accommodate large volumes of climate and epidemiological data. Modular components allow the system to expand as additional datasets or predictive models are incorporated. Cloud-based computing infrastructure or distributed processing environments can be used to handle computationally intensive deep learning tasks and support real-time forecasting.

Security considerations are also critical due to the sensitive nature of public health data. Access control mechanisms, data encryption, and secure data storage practices will be implemented to ensure confidentiality and integrity. Additionally, the system will follow data governance guidelines and institutional policies to maintain ethical data usage and prevent unauthorized access.

5. USER REQUIREMENTS

5.1. Stakeholder Identification

The proposed dengue risk forecasting system involves several stakeholders who interact with the system directly or benefit from its outputs.

Public Health Authorities such as the Sri Lankan Ministry of Health and the Epidemiology Unit are the primary stakeholders. They require early warnings and predictive insights to implement vector control strategies and allocate healthcare resources efficiently.

Local Government and Municipal Authorities are responsible for environmental management and sanitation activities. They can use dengue risk predictions to conduct targeted mosquito control operations in high-risk regions.

Researchers and Data Scientists involved in climate analytics and epidemiological studies may utilize the system for analyzing climate–disease relationships and improving predictive models.

General Public and Community Health Workers indirectly benefit from the system through early warning alerts and awareness campaigns that help reduce exposure to dengue risk.

Disaster Management and Policy Makers can also use predictive insights to integrate climate-driven health risk forecasting into national disaster preparedness strategies.

5.2. Functional Requirements

The system must support several core functional capabilities to ensure effective dengue risk forecasting and decision support.

FR1: The system shall collect climate data including rainfall, temperature, and humidity from meteorological data repositories.

FR2: The system shall collect historical dengue case records from official epidemiological databases.

FR3: The system shall preprocess and integrate heterogeneous datasets using data cleaning, normalization, and spatiotemporal alignment techniques.

FR4: The system shall generate climate-based predictive features such as lagged rainfall indicators and temperature anomalies.

FR5: The system shall train and deploy ML and deep learning models including LSTM and Transformer-based architectures for dengue risk prediction.

FR6: The system shall generate weekly dengue risk predictions for different geographic regions in Sri Lanka.

FR7: The system shall produce spatial risk maps and hotspot identification outputs for visualization.

FR8: The system shall provide decision support outputs to assist health authorities in implementing preventive interventions.

5.3. Non-Functional Requirements

Non-functional requirements describe the quality attributes that ensure the system operates effectively, securely, and reliably.

Security

The system must implement secure data storage and controlled access mechanisms to protect sensitive epidemiological data. Data encryption and authentication protocols should be applied to prevent unauthorized access.

Performance

The system should process large climate and epidemiological datasets efficiently and generate dengue risk predictions within acceptable time limits. Model training and prediction processes should support scalable computation.

Scalability

The system architecture should allow integration of additional datasets, geographic regions, or predictive models without requiring major system redesign.

Usability

The visualization interface should be user-friendly and accessible to non-technical stakeholders such as public health officials. Risk maps and dashboards should present information clearly and support quick decision-making.

Reliability

The system should provide consistent predictive outputs and maintain stable performance during continuous data processing and model execution.

Maintainability

The system should be modular so that individual components such as data pipelines, predictive models, or visualization modules can be updated independently.

6. COMMERCIALIZATION PLAN

6.1. Target Market and Customer Persona

The proposed AI-driven dengue risk forecasting system primarily targets institutions and organizations involved in public health management, environmental monitoring, and urban planning in Sri Lanka.

The primary customers are government health authorities such as the Ministry of Health, the Epidemiology Unit, and the National Dengue Control Unit, which are responsible for monitoring dengue outbreaks and implementing prevention programs. These institutions require reliable forecasting tools to support early warning systems, vector control planning, and healthcare resource allocation.

The secondary customers include organizations that require climate-related health risk insights for planning and risk management. These organizations may include municipal councils, disaster management agencies, research institutions, insurance companies, and urban development authorities. Access to predictive dengue risk analytics can help these stakeholders make informed decisions about infrastructure planning, environmental management, and public safety.

The general public and community health workers represent an additional group of end users who benefit indirectly from the system. Through web dashboards or mobile applications, citizens can access dengue risk alerts and hotspot maps that provide early warnings about potential outbreaks in their areas. This information can encourage preventive actions such as eliminating mosquito breeding sites and seeking early medical attention.

6.2. Value Proposition

The proposed system provides a data-driven dengue risk forecasting platform that integrates climate data and epidemiological records to generate predictive insights for public health decision-making.

Unlike traditional dengue monitoring systems that rely mainly on retrospective case reporting, the proposed platform uses artificial intelligence and climate analytics to predict potential dengue outbreaks before they occur. By analyzing rainfall, temperature, humidity, and historical dengue data, the system can identify environmental patterns that increase the risk of mosquito breeding and disease transmission.

For government health authorities, the platform enables proactive intervention by identifying high-risk areas and supporting targeted vector control strategies. For organizations and researchers, the system provides valuable climate–health insights that can support environmental planning and risk assessment. For the general public, the system provides accessible dengue risk alerts and visualizations that help communities respond more effectively to disease threats.

6.3. Revenue Model

The commercialization strategy focuses primarily on institutional licensing and service-based deployment.

The main revenue streams include:

Government system licensing - National and regional health authorities can license the system for official dengue monitoring and public health planning.

Enterprise licensing - Research institutions, insurance companies, environmental consultancies, and urban planning agencies may subscribe to the platform for access to predictive analytics and risk assessments.

Data analytics services - Customized climate–health analysis reports can be provided to organizations conducting infrastructure development or environmental impact studies.

Public access services - Basic dengue alerts and risk maps will remain freely available to the public. However, advanced analytical dashboards and historical data insights may be offered through a subscription-based service for professional users.

This model ensures that essential public health information remains freely accessible while maintaining financial sustainability through institutional partnerships.

6.4. Cost Estimation

The major operational costs involve cloud computing infrastructure, data storage, system maintenance, and technical support. Development Costs

Table 1: Cost Estimation

Cost Component	Estimated Monthly Cost (LKR)	Annual Cost (LKR)
Cloud computing for model training and forecasting	7,000	84,000
Database and data storage services	3,500	42,000
Data backup and disaster recovery	2,500	30,000
Security and encryption services	1,500	18,000
System maintenance and monitoring	35,000	420,000

6.5. Pricing Strategy

The pricing model prioritizes institutional deployment while ensuring public accessibility to essential dengue warning services.

Table 2 : Pricing Strategy

Service Type	Price (LKR)	Description
Government institutional license	1,800,000	Full system deployment, training, and integration
Enterprise organization license	1,400,000	Access to dengue risk analytics platform
Advanced climate–health analysis reports	180,000 per report	Custom outbreak risk assessment
Public web/mobile access	Free	Basic dengue alerts and hotspot maps

6.6. Competitive Advantage

The proposed system provides several advantages compared with traditional dengue monitoring approaches:

AI-driven predictive forecasting identifies potential dengue outbreaks before they occur.

Integration of climate and epidemiological data, enabling more accurate modeling of environmental disease dynamics.

High-resolution dengue risk maps that allow authorities to identify localized outbreak hotspots.

Scalable cloud-based architecture capable of processing large datasets and supporting real-time forecasting.

Decision support capabilities that help public health authorities implement targeted intervention strategies.

These features make the system significantly more effective than conventional surveillance tools that rely only on historical disease reporting.

6.7. Estimated Project Budget

The development of the proposed AI-driven dengue risk forecasting system requires several technical and operational resources including computing hardware, cloud infrastructure, software tools, and human effort. These resources support activities such as data preprocessing, ML model training, system development, and visualization of predictive outputs.

6.8. Intellectual Property Considerations

The intellectual property associated with the proposed system includes the AI-based dengue forecasting algorithms, spatiotemporal data processing pipeline, and system architecture used for predictive modeling and visualization.

The developed software platform can be protected through software copyright and licensing agreements, ensuring that the predictive models and system components remain the intellectual property of the developers.

Institutional deployments will operate under formal licensing contracts, allowing authorized organizations to use the system while protecting the underlying software and algorithms. Additionally, all external datasets used within the system will comply with data sharing policies and access regulations established by relevant government agencies and research institutions.

These intellectual property protections ensure that the system can be commercially deployed while maintaining ownership and control over the developed technologies.

7. BUDGET AND JUSTIFICATION

7.1. Estimated Project Budget

This section presents the estimated budget required for the development and evaluation of the AI-driven dengue risk forecasting system. Since this research is conducted as a student-level academic project, the overall budget is minimized by utilizing open-source software, existing personal hardware, and free-tier cloud services where possible. Paid services are included only when necessary for computational resources and deployment support.

7.2 Hardware Costs

The research will primarily be conducted using an existing personal laptop owned by the researcher. Therefore, no additional hardware purchases are required.

Table 3 : Hardware Cost

Hardware	Purpose	Cost (LKR)
Personal Laptop	Data preprocessing, model development, visualization	0
External Storage (Optional)	Backup of datasets and model outputs	5,000
TOTAL		5,000

The use of an existing laptop significantly reduces project costs. Optional external storage is allocated for safe backup of climate datasets and trained model outputs.

7.3 Software Licenses

The system development primarily relies on free and open-source software tools widely used in data science and ML research.

Table 4 : Software Cost

Software	Purpose	License Type	Cost (LKR)
Python 3.x	Data processing and system development	Free / Open Source	0

TensorFlow	Deep learning model development (LSTM, Transformers)	Free / Open Source	0
Scikit-learn	Baseline ML models	Free / Open Source	0
VS Code	Integrated development environment	Free	0
Google Colab Pro	GPU-accelerated model training	Paid (annual)	25,000
TOTAL			25,000

Python and TensorFlow provide powerful ML capabilities without licensing costs. Google Colab Pro is included to provide GPU resources required for training deep learning models efficiently, which would otherwise require expensive hardware.

7.4 Cloud Services

Cloud services may be required for data storage, model hosting, and visualization dashboards.

Table 5 : Cloud Services Cost

Service	Usage	Plan	Cost (LKR)
Google Cloud Storage	Dataset storage and backup	Free tier	0
Firebase	Dashboard backend and authentication	Free tier	0
Google Maps API	Visualization of dengue risk maps	Free tier	0
Cloud Compute (Optional)	Model inference and deployment	Pay-as-you-go	5000
TOTAL			5000

The free tiers of Google Cloud and Firebase are sufficient for a student prototype. A small contingency cost is included for optional cloud computing if additional model inference resources are required.

7.5 Human Effort Cost

Since this project is conducted by a student researcher, there are no direct labor costs. However, minor expenses are expected for report preparation and submission.

Table 6 : Human Effort Cost

Role	Task	Cost (LKR)
Printing & Binding	Research report and documentation submission	2,000
TOTAL		2,000

7.6 Justification of Expenses

Table 7 : Project Budget Summary

Category	Total Cost (LKR)
Hardware Costs	5,000
Software Licenses	25,000
Cloud Services	5,000
Human Effort Cost	2,000
GRAND TOTAL	LKR 37,000

The estimated project budget reflects the essential resources required to successfully develop and evaluate the AI-driven dengue forecasting system. Hardware resources provide the computational environment needed for ML experimentation and software development. Cloud computing services enable scalable model training and data storage, which is particularly important when working with large climate datasets.

The use of open-source software technologies significantly reduces licensing expenses while maintaining access to state-of-the-art ML tools. Meanwhile, the allocation for human effort represents the research, technical expertise, and development time invested in building the predictive system.

Overall, the proposed budget ensures that the project remains technically feasible, cost-effective, and suitable for an academic research environment while supporting the successful implementation of the proposed AI-based forecasting solution.

8. WORK BREAKDOWN STRUCTURE (WBS)

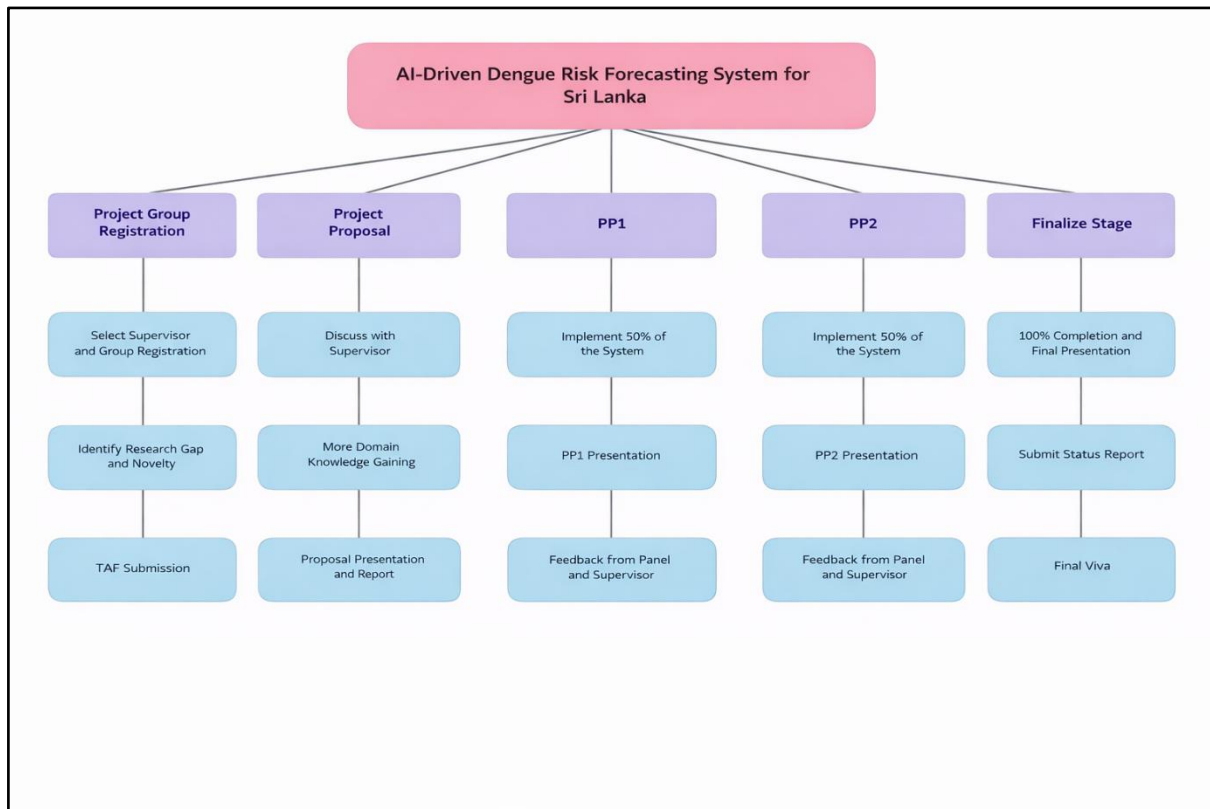


Figure 3 : Work Breakdown Structure

9. GANTT CHART

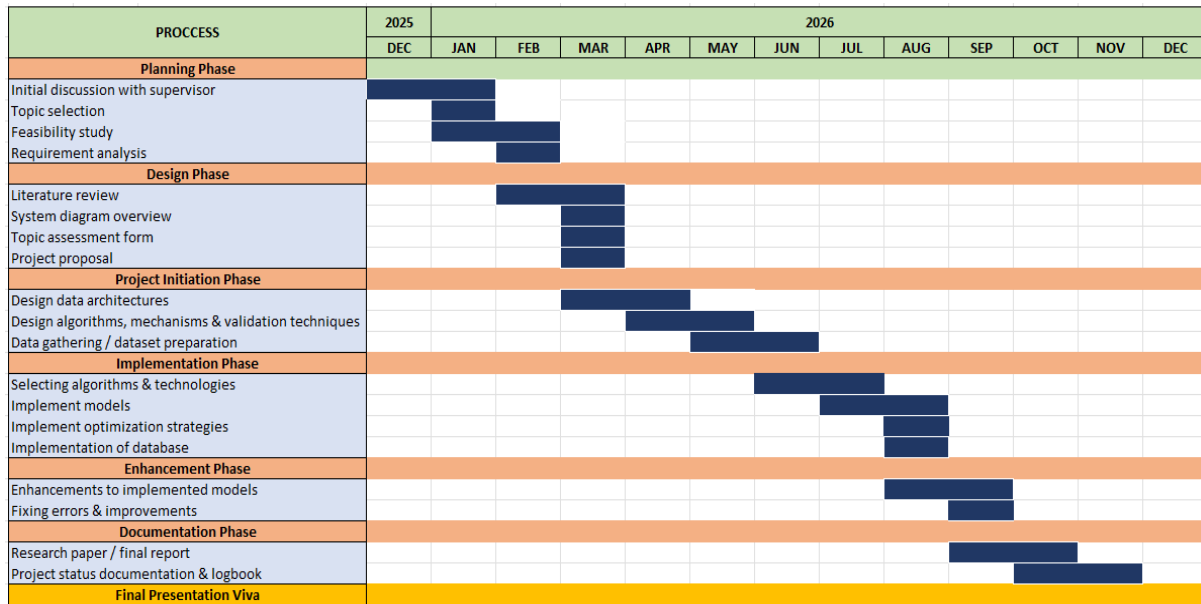


Figure 4 : Grantt Chart

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11. APENDICEIS

Appendix A - The number of reported dengue cases between 1950 – 2017

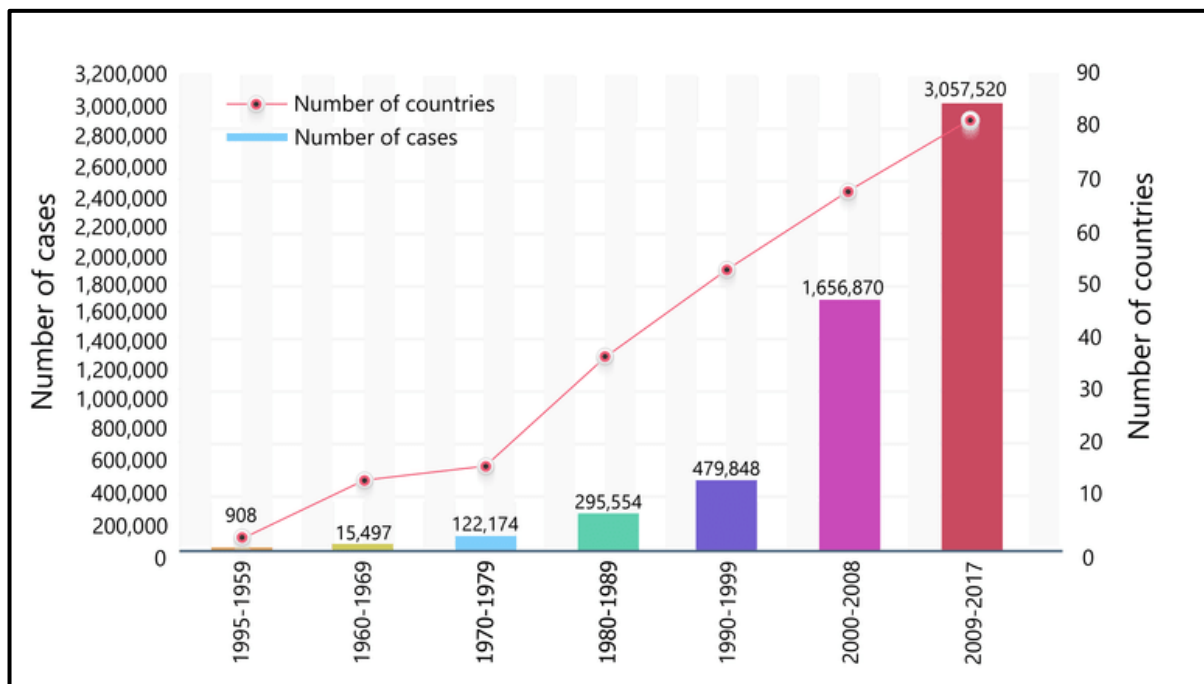


Figure 5 : The Number Of Reported Dengue Cases Between 1950 - 2017

Appendix B - 2006 - 2017 dengue spreading map

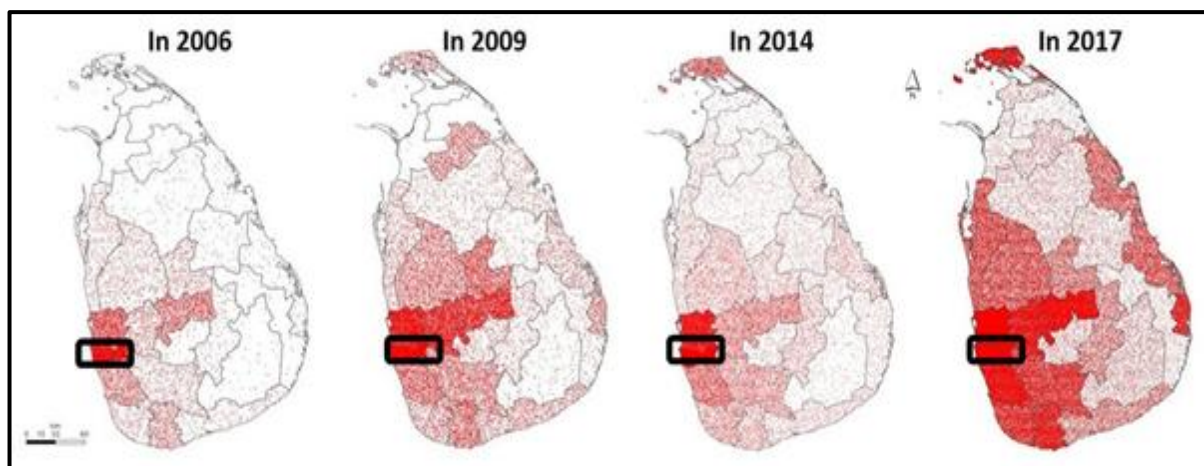


Figure 6 : 2006 - 2017 dengue spreading map

AI-Driven Climate-Aware Decision Support and Risk Forecasting Systems for Sri Lanka

R26-IT-046

Project Proposal Report

Athukorala C.J – IT22299666

B.Sc. (Hons) Degree in Information Technology Specialized in
Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka

March 2026

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Agriculture plays a vital role in ensuring food security and supporting rural livelihoods in Sri Lanka, where paddy cultivation is a major agricultural activity. However, increasing climate variability, including unpredictable rainfall patterns and temperature fluctuations, has significantly affected paddy productivity and farming decisions. Many farmers still rely on traditional knowledge and general weather forecasts when planning irrigation, fertilizer application, and other crop management activities. These approaches often lack data-driven insights, which may lead to inefficient resource usage and reduced crop yields under changing climatic conditions.

Although recent research has explored the use of climate data analytics and machine learning in agriculture, most studies mainly focus on crop yield prediction rather than providing practical advisory recommendations that farmers can directly apply in their cultivation practices. In addition, many agricultural monitoring systems lack integrated decision-support features specifically designed for paddy farming. This highlights a research gap in developing systems that transform climate data into actionable farming guidance.

To address this challenge, this research proposes a Climate-Informed Paddy Advisory System that integrates historical climate data, crop growth stage information, and agricultural datasets to generate climate-based recommendations. The system analyzes variables such as rainfall and temperature to provide advisory insights related to irrigation scheduling, fertilizer application timing, and crop management practices. Technologies such as Python, TensorFlow, Firebase, and React Native will be used to implement the system.

The expected outcome is a practical advisory platform that helps farmers make informed cultivation decisions, improve resource management, and enhance climate resilience in paddy farming.

Keywords: *Climate-Informed Agriculture, Paddy Cultivation, Machine Learning, Decision Support Systems, Smart Farming*

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Finally, I would like to acknowledge the future contributions of all survey participants and stakeholders whose insights and cooperation will be valuable for the successful completion of this research. Their participation will play an important role in supporting the findings and practical implementation of this study.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
ML	Machine Learning
MAE	Mean Absolute Error
RMSE	Root Mean Squared Error
AI	Artificial Intelligence
GIS	Geographic Information Systems
IT	Information Technology
FR	Functional Requirement
SDG	Sustainable Development Goal

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1. INTRODUCTION

1.1 Background and Context

Paddy cultivation plays a vital role in the agricultural sector of Sri Lanka and contributes significantly to the country's food security and the economic livelihood of farmers. A large proportion of farmers in Sri Lanka depend on paddy cultivation as their primary source of income and livelihood. According to the Department of Census and Statistics Sri Lanka, rice is the staple food of the country, and paddy farming occupies a major share of the agricultural land area. Sri Lanka produces several million metric tons of paddy annually, making it one of the most important crops for national food security [2].

However, in recent years, climate variability and climate change have created significant challenges for paddy cultivation. Irregular rainfall patterns, temperature fluctuations, prolonged droughts, and sudden floods have increasingly affected agricultural productivity [2], [3]. These climate-related changes disrupt traditional cultivation schedules and create difficulties in water management, irrigation planning, and fertilizer application [1], [6]. Such climate-related challenges can lead to reduced paddy yields, decreased farmer income, inefficient use of water and fertilizer resources, and potential risks to national food security [5], [9]. Small-scale farmers are particularly vulnerable to these impacts because their livelihoods heavily depend on successful harvests. Statistical studies show that variations in rainfall and temperature have a direct influence on rice yield in many parts of South Asia, including Sri Lanka [2], [7].

Currently, many farmers still rely on traditional farming knowledge (TFK) or general weather forecasts when making cultivation decisions. Although these approaches provide basic guidance, they are not data-driven and often fail to deliver precise, location-specific recommendations [4], [11]. As a result, farmers face difficulties in predicting rainfall patterns, determining appropriate irrigation schedules, and applying fertilizers at the correct time. These limitations increase the risk of crop loss and reduce overall agricultural productivity [1], [5].

To address these challenges, this research proposes the development of a Climate-Informed Paddy Advisory System (CIPAS). The proposed system integrates historical paddy production data with climate variables such as rainfall and temperature to analyze the impact of climate patterns on crop performance. By utilizing data-driven techniques, including machine learning models, the system can generate advisory recommendations to support farmers in making better cultivation decisions [1], [3], [7]. Through this system, farmers can receive guidance related to irrigation scheduling, fertilizer application timing, and other cultivation management practices. The implementation of such a system has the potential to improve paddy yield, reduce climate-related risks, and promote the efficient use of agricultural resources [5], [10].

Furthermore, this research aligns with global sustainable development initiatives, particularly **Sustainable Development Goal (SDG) 2 – Zero Hunger** and **SDG 13 – Climate Action** [10]. From a technological perspective, this research falls within the **Centre of Excellence in Artificial Intelligence (CoEAI) research cluster**, where advanced data analytics and machine learning techniques are applied to address real-world agricultural challenges and support intelligent decision-making in farming systems, particularly within the domains of

Artificial Intelligence in Agriculture (AI4Ag) and Climate-Smart Agriculture (CSA) [4], [7].



Figure 1 Impact of natural disasters on the efficiency of agricultural production

1.2 Literature Review

Climate variability and climate change have become significant challenges for agricultural productivity worldwide, especially for climate-sensitive crops such as paddy rice. Variations in rainfall patterns, temperature fluctuations, and extreme weather events directly affect crop growth stages and yield outcomes. In countries such as Sri Lanka, paddy cultivation plays a crucial role in ensuring national food security and supporting the livelihoods of a large proportion of rural farmers. However, many farmers still rely on traditional knowledge and general weather forecasts when making important farming decisions such as irrigation scheduling, fertilizer application, and crop management. These methods are often not data-driven and therefore may lead to inefficient resource usage and reduced crop yields under changing climatic conditions.

To address these challenges, the concept of a **Climate-Informed Paddy Advisory System** has emerged as a promising approach for improving agricultural decision-making. Such systems integrate historical climate data, agricultural datasets, and predictive analytics to generate data-driven recommendations for farmers. By analyzing variables such as rainfall, temperature, and historical yield patterns, the system can support farmers in determining optimal irrigation schedules, fertilizer application timing, and crop management strategies. The main objective of this approach is to transform raw climate and agricultural data into actionable insights that help farmers reduce climate risks, improve productivity, and utilize resources more efficiently.

Recent research has explored various approaches to integrating climate information with agricultural decision support systems. Several studies have investigated crop yield prediction

using machine learning models that utilize climate and soil datasets [1], [3]. These approaches demonstrate that algorithms such as Random Forest and Support Vector Machines can predict crop yield variation more accurately than traditional statistical models. However, most of these studies focus primarily on yield prediction and do not provide practical advisory recommendations that farmers can directly apply in their cultivation practices.

Climate variability and its influence on agricultural productivity have also been widely analyzed in global studies [2]. These studies emphasize the importance of incorporating weather and climate data into agricultural planning and management. Similarly, research on Artificial Intelligence applications in agriculture highlights the potential of AI technologies for crop monitoring, yield estimation, and data analysis [4]. Despite these advancements, many AI-based systems still lack integrated advisory components that can support farmers in making real-time cultivation decisions.

Decision support systems have also been developed to assist farmers in irrigation management and crop planning. Some systems utilize weather forecasts and crop growth models to optimize irrigation scheduling [6]. While these systems improve water management, they often focus only on irrigation and do not include additional advisory functions such as fertilizer application timing or crop stage management. In addition, agricultural advisory services provided by international organizations emphasize the importance of climate-smart farming practices and information dissemination to farmers [10].

Furthermore, several studies have explored the use of machine learning techniques and climate datasets for crop yield analysis and agricultural prediction models [7], [9]. These models demonstrate that integrating climate variables with agricultural datasets can significantly improve prediction accuracy. However, many of these studies remain focused on analytical models rather than implementing practical decision support tools that farmers can easily access.

Although commercial smart farming platforms provide integrated weather monitoring and agricultural analytics, these systems are often designed for large-scale commercial farming operations and may not be accessible or affordable for smallholder farmers [5], [11]. Additionally, such platforms often require complex technological infrastructure and technical expertise, which limits their usability in developing agricultural communities.

Therefore, while existing studies demonstrate the potential of climate data analytics and machine learning in agriculture, many current systems still lack integrated advisory features tailored for specific crops such as paddy. This highlights the need for a climate-informed paddy advisory system that combines historical yield data, climate variables, and predictive analytics to provide practical farming recommendations and support data-driven agricultural decision-making

1.3 Research Gap

Despite the growing body of research on climate data analytics and machine learning applications in agriculture, several important gaps remain in the development of practical decision-support tools for farmers. Many existing studies primarily focus on crop yield prediction using climate and soil datasets rather than providing actionable recommendations that farmers can directly apply in their daily farming activities [1], [3], [7]. While predictive models are capable of identifying patterns between climate variables and crop productivity, these models often remain as analytical tools and are not translated into advisory systems that support real-time agricultural decision-making.

In addition, existing agricultural decision-support systems mainly concentrate on specific aspects of farm management such as irrigation optimization or yield estimation [6]. Although these systems improve certain farming practices, they typically do not integrate multiple advisory functions such as irrigation scheduling, fertilizer application timing, and crop stage management within a single platform. As a result, farmers must rely on multiple sources of information, which limits the practicality and efficiency of these systems.

Another limitation identified in current agricultural technologies is the accessibility and usability of existing smart farming platforms. Many commercial agricultural analytics platforms are designed primarily for large-scale commercial farms and often require advanced technological infrastructure, specialized knowledge, and high operational costs [5], [11]. This makes such systems difficult to adopt for smallholder farmers, particularly in developing agricultural regions such as Sri Lanka, where farmers may have limited access to advanced digital tools and data-driven decision support systems.

Furthermore, many existing studies focus primarily on prediction accuracy and algorithm performance rather than on developing integrated systems that transform climate data into practical farming recommendations. This highlights a lack of user-centered agricultural advisory platforms that combine historical climate data, crop yield records, and predictive analytics to support real-world farming decisions.

Therefore, there is a clear research gap in the development of an integrated **Climate-Informed Paddy Advisory System** that can analyze climate variables, historical yield data, and agricultural practices to generate practical recommendations for irrigation scheduling, fertilizer application timing, and crop management. By transforming climate data into actionable insights, such a system can help farmers make more informed decisions, reduce climate-related risks, and improve agricultural productivity. The proposed research addresses this gap by developing a data-driven advisory platform specifically tailored for paddy cultivation, thereby contributing a novel and practical solution for climate-informed agricultural decision-making

2. OBJECTIVE

2.1 Main Objective

To develop and evaluate a **Climate-Informed Paddy Advisory System** that integrates historical climate data, crop yield data, and predictive analytics to generate data-driven recommendations for irrigation scheduling, fertilizer application timing, and crop management, in order to support farmers in making informed cultivation decisions and improving paddy productivity

2.2 Specific Objectives

- To collect and preprocess historical climate data, paddy yield records, and relevant agricultural datasets to create a structured dataset for analysis and model development.
- To analyze the relationship between climate variables (such as rainfall and temperature) and paddy yield patterns in order to identify key factors affecting crop productivity.
- To develop a predictive model using machine learning techniques to estimate paddy yield and cultivation conditions based on historical climate and agricultural data.
- To design and implement an advisory module that generates recommendations for irrigation scheduling and fertilizer application timing based on climate-informed insights.
- To develop a user-friendly digital platform that allows farmers to access climate-based advisory information and visualize relevant agricultural insights.
- To evaluate the effectiveness of the proposed system by testing prediction accuracy and assessing the usefulness of generated recommendations for supporting farmer decision-making.

3. METHODOLOGY

3.1 Research approach

This research follows a **Design Science approach**, focusing on creating a practical, data-driven advisory system for paddy farmers in Sri Lanka. Design Science is appropriate because the project involves **developing an artifact**—the Climate-Informed Paddy Advisory System—that collects, processes, and analyzes climate and historical yield data to generate **personalized recommendations for irrigation scheduling and fertilizer application**. The approach emphasizes **iterative design, implementation, and evaluation**, ensuring that the system is technically robust, scalable, and usable by farmers and agricultural extension officers

3.2 Technical Methods and Algorithms

The core predictive model of the Climate-Informed Paddy Advisory System employs a **hybrid architecture combining Long Short-Term Memory (LSTM) and Random Forest algorithms** to capture temporal climate patterns and agricultural decision relationships. The LSTM component analyzes time-series climate variables such as rainfall patterns, temperature variations, and seasonal trends that influence paddy growth stages. This component is implemented using TensorFlow due to its capability to efficiently process sequential climate data.

Historical datasets including paddy yield records, rainfall, temperature, soil characteristics, and other climatic variables are preprocessed through data cleaning, normalization, and temporal alignment. Missing values are handled using appropriate imputation techniques to ensure reliable inputs for model training.

After preprocessing, the LSTM model extracts temporal climate features and patterns. These features are then combined with crop and soil variables and provided as inputs to a Random Forest model implemented using scikit-learn. The Random Forest algorithm identifies complex relationships between environmental conditions and crop productivity to predict yield variations and cultivation conditions. The generated predictions are converted into actionable recommendations through a decision support module, providing farmers with guidance on irrigation scheduling and fertilizer application timing, supporting climate-informed agricultural decision making.

3.3 Tools and Platforms with Justification

The Climate-Informed Paddy Advisory System is implemented using a combination of **Python, TensorFlow, Firebase, React Native, and GIS tools**, selected specifically to meet the technical requirements of the project while ensuring accessibility for smallholder farmers.

Python is employed for data processing, cleaning, normalization, feature extraction, and integration of climate and yield datasets. Its extensive libraries such as Pandas, NumPy, and SciPy allow efficient manipulation of large datasets, while visualization libraries like Matplotlib and Seaborn help in exploratory data analysis. Python was chosen over other languages due to its flexibility, community support, and strong ecosystem for scientific computing.

TensorFlow is used to develop and train the LSTM component of the predictive model due to its stability, GPU support, and ability to efficiently process sequential climate datasets. Alternative frameworks such as PyTorch were considered; however, TensorFlow provides smoother integration with cloud deployment services and easier model serialization for mobile applications. A Random Forest model is additionally used to analyze combined climate, soil, and crop features to generate practical farming recommendations.

Firestore serves as the cloud backend, providing real-time database management, cloud storage, and user authentication. Firestore was chosen over traditional SQL or NoSQL databases because it supports seamless synchronization between the mobile application and backend, enabling timely updates of advisory recommendations based on the latest climate and yield predictions. Its built-in security rules ensure that sensitive user and location data are protected.

React Native is used to develop the mobile application, allowing a single codebase to be used for the application development process. Native development using languages such as Java, Kotlin, or Swift was not selected because it would significantly increase development time and maintenance overhead. React Native also integrates directly with Firestore, simplifying the data flow from the backend to the user interface and enabling interactive features such as advisory dashboards and maps.

GIS tools are incorporated to visualize spatial climate patterns and paddy cultivation areas. By overlaying rainfall, temperature, and yield data on geographic maps, farmers can understand regional climate trends and make informed location-specific decisions. Together, these tools create a robust, efficient, and user-friendly system capable of processing large datasets, generating accurate predictive insights, and delivering actionable advisories to farmers. The chosen combination balances technical capability with accessibility, ensuring that the system is both effective and practical for smallholder paddy farmers in Sri Lanka.

3.4 Data Collection and Ethical Considerations

Data sources include historical paddy yield records, district-level climate data (rainfall, temperature), and other relevant agricultural datasets obtained from official Sri Lankan agencies. Python pipelines automate data collection, cleaning, normalization, and temporal alignment to produce structured datasets for modeling. Ethical considerations involve acknowledging data sources, ensuring privacy and security of location-specific data, and adhering to best research practices. Firestore security rules are implemented to protect user data and prevent unauthorized access.

3.5 Validation Strategy and Performance Metrics

The system is validated by comparing predicted paddy yields and cultivation advisories against historical yield outcomes and real climate patterns. **Performance metrics** include **Mean Absolute Error (MAE)** and **Root Mean Squared Error (RMSE)** for yield predictions, alongside accuracy, precision, recall, and F1-score for classification of optimal irrigation or fertilization timing. Additionally, the usability and effectiveness of the mobile advisory platform are assessed through farmer feedback, measuring clarity, responsiveness, and practicality of recommendations. This combined evaluation ensures the system is accurate, reliable, and actionable for end users.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

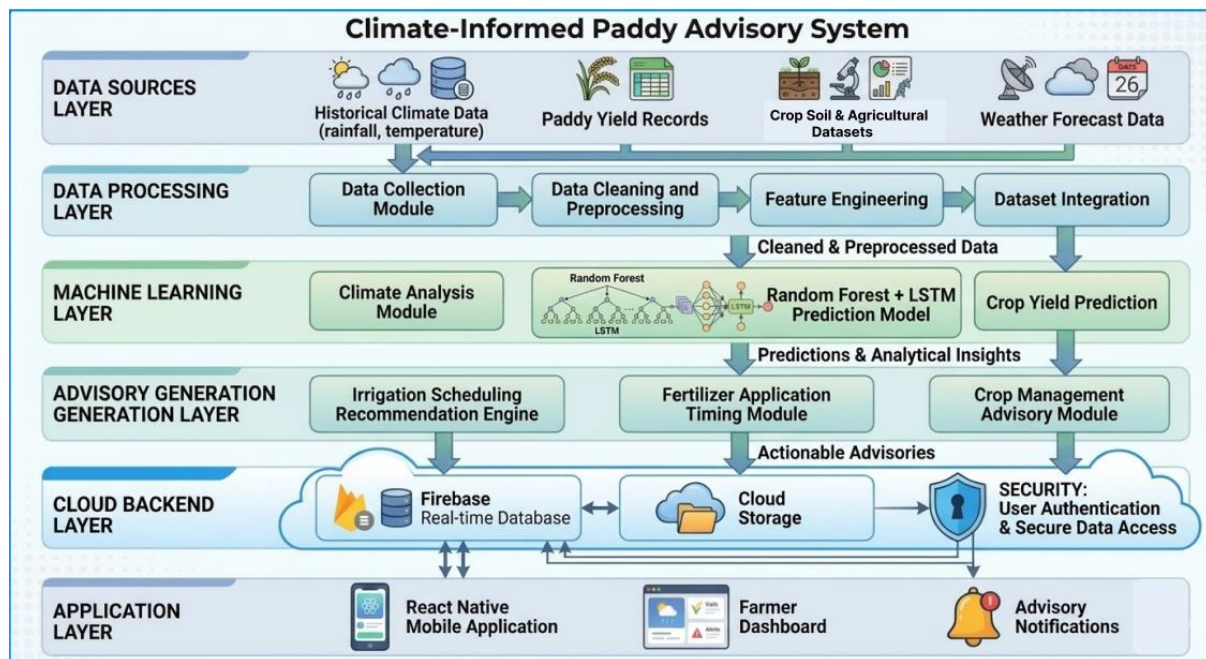


Fig.2 High level System Architecture

5. USER REQUIREMENTS

5.1 Stakeholder Identification

The proposed **Climate-Informed Paddy Advisory System** involves several stakeholders who interact with or benefit from the system.

1. Paddy Farmers (Primary Users)

Paddy farmers are the main users of the system. They use the mobile application to receive climate-informed recommendations related to irrigation scheduling, fertilizer application timing, and crop management. The system helps them make better farming decisions based on climate data and predictive insights.

2. Agricultural Experts / Agronomists

Agricultural experts can use the system to review recommendations generated by the model and provide domain knowledge to improve advisory accuracy. Their feedback can help validate the system's recommendations.

3. Agricultural Authorities

Government agricultural departments and extension services can use the system to monitor climate impacts on paddy cultivation and support farmers with more targeted advisory services.

4. System Administrators / Developers

System administrators are responsible for maintaining the backend infrastructure, managing data updates, monitoring system performance, and ensuring the reliability of the platform.

5.2 Functional Requirements

The functional requirements describe the core features that the system must provide.

FR1: The system shall collect and store historical climate data such as rainfall and temperature.

FR2: The system shall allow the processing and integration of climate and agricultural datasets for analysis.

FR3: The system shall use a machine learning model to analyze climate patterns and generate predictions related to paddy cultivation.

FR4: The system shall generate advisory recommendations for irrigation scheduling based on climate data analysis.

FR5: The system shall generate recommendations for fertilizer application timing based on crop growth stages and environmental conditions.

FR6: The system shall provide a mobile application interface where farmers can view advisory recommendations.

FR7: The system shall allow users to view relevant climate information and predictions through the mobile interface.

FR8: The system shall store generated recommendations in a cloud database for access by the mobile application.

5.3 Non-Functional Requirements

Non-functional requirements define the quality attributes and operational constraints of the system. These requirements ensure that the system is reliable, efficient, secure, and easy for farmers to use.

1. Security

The system must ensure that user data and system resources are protected from unauthorized access. Secure authentication mechanisms should be implemented to ensure that only authorized users can access the system. In addition, data stored in the cloud database should be protected using encryption and controlled access policies. Security is important because the system may store sensitive user information and agricultural data.

2. Performance

The system must process data and deliver recommendations within a reasonable time frame. Efficient data processing and optimized machine learning models are necessary to ensure that advisory recommendations are generated quickly. High system performance is important because farmers require timely information to make decisions related to irrigation and fertilizer application.

3. Usability

The mobile application must provide a simple and user-friendly interface so that farmers can easily understand and use the system. Since many farmers may have limited technical experience, the interface should include clear navigation, simple language, and visual indicators where necessary. Good usability ensures that farmers can effectively interpret and apply the recommendations generated by the system.

4. Scalability

The system should be capable of handling an increasing number of users and datasets as the platform expands. A scalable architecture ensures that the system can support more farmers and additional climate data sources without affecting performance.

5. Reliability

The system must operate consistently and provide accurate advisory recommendations based on the available data. Reliable system behavior ensures that farmers can trust the recommendations provided by the platform when making critical farming decisions.

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

The proposed **Climate-Informed Paddy Advisory System** targets three main user groups: farmers, agricultural institutions, and agricultural support organizations.

The primary users are **paddy farmers**, particularly small and medium-scale farmers who depend on rice cultivation as their main source of income. These farmers often face challenges related to unpredictable weather conditions, inefficient fertilizer usage, and improper irrigation scheduling. By providing climate-informed recommendations, the system helps farmers make better decisions regarding crop management, ultimately improving productivity and reducing losses.

The secondary users include **government agricultural agencies and extension services**, such as the **Department of Agriculture Sri Lanka** and regional agricultural offices. These institutions can use the platform to provide data-driven advisory services to farmers and improve agricultural planning and support programs.

Additionally, **agricultural organizations and research institutions** represent another important user group. Organizations involved in agricultural development, climate research, and rural development programs can use the system to analyze climate impacts on crop production and develop better agricultural policies.

6.2 Value Proposition

The proposed system provides a **climate-informed agricultural advisory platform** that integrates climate data, crop growth stage information, and historical paddy yield data to generate intelligent farming recommendations.

Unlike traditional farming advisory approaches that rely mainly on general weather forecasts or static crop calendars, the proposed system combines **climate analysis and machine learning techniques** to generate data-driven recommendations for irrigation scheduling, fertilizer application timing, and crop management.

For farmers, the system helps improve crop productivity, reduce unnecessary fertilizer usage, and manage climate-related risks such as drought or excessive rainfall. For agricultural institutions, the platform supports better agricultural planning and improves the effectiveness of extension services. Overall, the system promotes **more efficient, climate-resilient, and sustainable farming practices**.

6.3 Revenue Model

The system can generate revenue through a combination of institutional partnerships and agricultural service subscriptions.

Potential revenue streams include:

Government agricultural licensing – Government agricultural institutions can adopt the platform to support national agricultural advisory services and farmer support programs.

Agricultural organization partnerships – Development organizations, agricultural research institutions, and climate adaptation programs may use the system for agricultural monitoring and advisory services.

Premium advisory services – While basic farming recommendations will remain free for farmers, advanced analytics such as seasonal yield prediction, climate risk analysis, or customized farm advisory reports may be offered through a small subscription fee.

This approach ensures that essential farming advice remains accessible to farmers while generating revenue through institutional collaborations.

6.4 Cost Estimation

The main operational costs of the system include cloud infrastructure, database services, and system maintenance.

Table 1 Cost Estimation

Cost Component	Estimated Monthly Cost (LKR)	Annual Cost (LKR)
Cloud computing for data processing	5,000	60,000
Database and storage services	2,500	30,000
Data storage and backup	2,000	24,000
Security and system monitoring	1,500	18,000
System maintenance and updates	25,000	300,000

Total Estimated Annual Operational Cost: ≈ LKR 432,000

These costs mainly cover the infrastructure required to maintain the data processing pipelines, climate data integration, and mobile application services

6.5 Pricing Strategy

The pricing strategy focuses on providing free access for farmers while generating revenue through institutional usage and advanced services.

Table II Pricing Strategy

Service Type	Price (LKR)	Description
Government institutional license	1,200,000	Full system deployment for agricultural advisory programs
Agricultural organization license	900,000	Access to advisory analytics and agricultural data insights
Advanced agricultural advisory reports	100,000 per report	Custom analysis for agricultural planning
Farmer mobile application	Free	Basic climate-informed advisory recommendations

This strategy ensures that small-scale farmers can access advisory services without financial barriers while maintaining system sustainability through institutional support.

6.6 Competitive Advantage

The proposed system offers several advantages compared with traditional agricultural advisory approaches:

- Integration of **climate data, crop growth stage information, and historical yield data**
- **Data-driven advisory recommendations** for fertilizer and irrigation scheduling
- Mobile application access that allows farmers to receive recommendations easily
- Ability to support **climate-resilient agriculture and sustainable farming practices**
- Scalable cloud-based infrastructure that can support multiple agricultural regions

These features make the system useful for both individual farmers and agricultural institutions working to improve agricultural productivity.

6.7 Intellectual Property Considerations

The intellectual property associated with the proposed system includes the **machine learning models, advisory algorithms, data integration pipelines, and system architecture** used to generate climate-informed farming recommendations.

The software implementation and predictive models can be protected under **software copyright and licensing agreements**. Institutional deployments of the system will operate under formal licensing contracts that define the rights and responsibilities of both developers and users.

All external datasets used within the system, including climate and agricultural statistics, will comply with the data access policies of the relevant organizations such as the **Department of Agriculture Sri Lanka** and other agricultural data providers.

7. BUDGET AND JUSTIFICATION

This section outlines the estimated budget required for the successful development and deployment of the **Climate-Informed Paddy Advisory System**. As a student-level research project, the budget is kept minimal by utilizing free and open-source tools, existing personal hardware, and free-tier cloud services whenever possible. This approach ensures that the system can be developed within a limited budget while still meeting the technical requirements of the project.

7.1 Hardware Cost

The researcher will use an existing personal laptop for data processing, machine learning model development, and mobile application development. Therefore, no additional hardware purchases are required for this project.

7.2 Software Licenses

Table III Software Licenses

Software	Purpose	License Type	Cost (LKR)
Python 3.x	Data processing and machine learning development	Free / Open Source	0
TensorFlow	Machine learning model training	Free / Open Source	0
React Native	Mobile application development	Free / Open Source	0
Visual Studio Code	Code editor for development	Free	0
Google Colab Pro	GPU-based model training	Paid (monthly)	25,000
TOTAL			25,000

All core development tools such as Python, TensorFlow, and React Native are free and open-source, making them suitable for a student-level research project. Google Colab Pro is used as a cost-effective alternative to purchasing expensive GPU hardware, enabling faster machine learning model training.

7.3 Cloud Services

Table VI Cloud Services

Service	Usage	Plan	Cost (LKR)
Firebase Spark Plan	Database, authentication, and data storage	Free tier	0
Firebase Blaze Plan	Cloud functions and data processing (if needed)	Pay-as-you-go	5,000
Weather Data APIs	Climate data access	Free tier	0
TOTAL			5,000

Firebase's free Spark Plan provides sufficient cloud infrastructure for the system, including real-time database management, authentication, and storage of advisory data. The Blaze Plan is budgeted as a contingency in case additional cloud processing or functions are required. Weather data APIs provide access to rainfall and temperature datasets necessary for climate-based analysis.

7.4 Human Effort Cost

Table V Human Effort Cost

Role	Task	Cost (LKR)
Printing & Binding	Research reports and documentation submission	2,000
TOTAL		2,000

Printing and binding costs are estimated for preparing and submitting research documents to the Department of Information Technology.

7.5 Budget Summary

TABLE VI Budget Summary

Category	Total Cost (LKR)
7.1 Hardware Costs	0
7.2 Software Licenses	25,000
7.3 Cloud Services	5,000
7.4 Human Effort Cost	2,000
GRAND TOTAL	LKR 32,000

The total estimated budget for this research project is **LKR 32,000**. This budget is relatively low because the project relies heavily on open-source software tools, existing personal hardware, and free-tier cloud services. The use of cost-effective cloud computing solutions such as Google Colab Pro allows machine learning models to be trained without requiring expensive hardware. This budget structure reflects the student-level scope of the research while ensuring that the necessary technologies are available to develop a functional climate-informed advisory system.

8. WORK BREAKDOWN STRUCTURE (WBS)

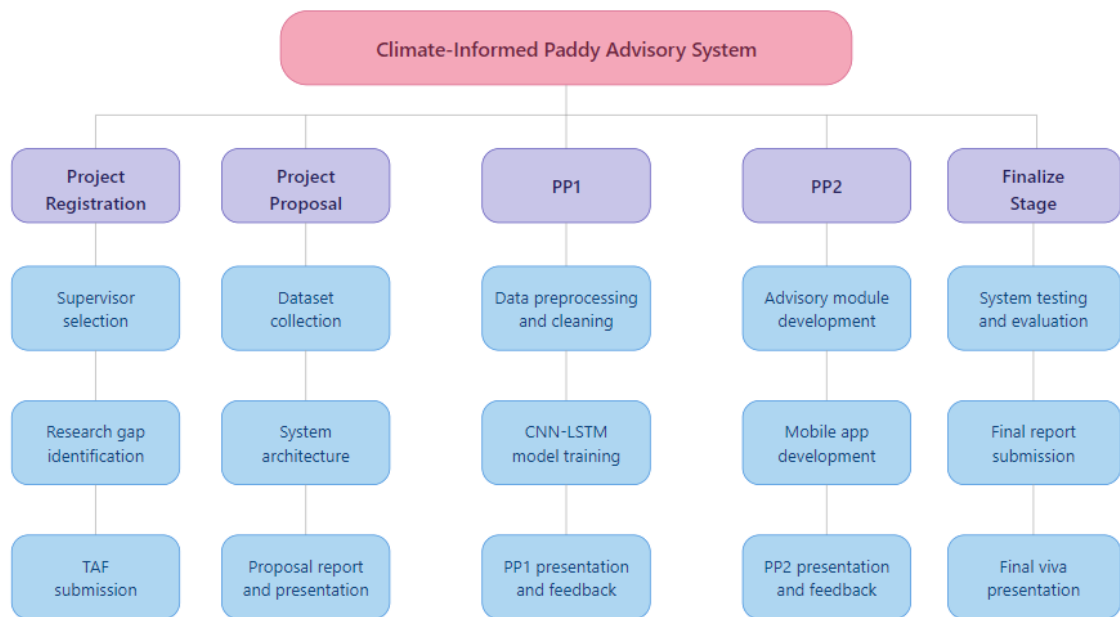


Fig.3 Work Breakdown Structure

9. GANTT CHART

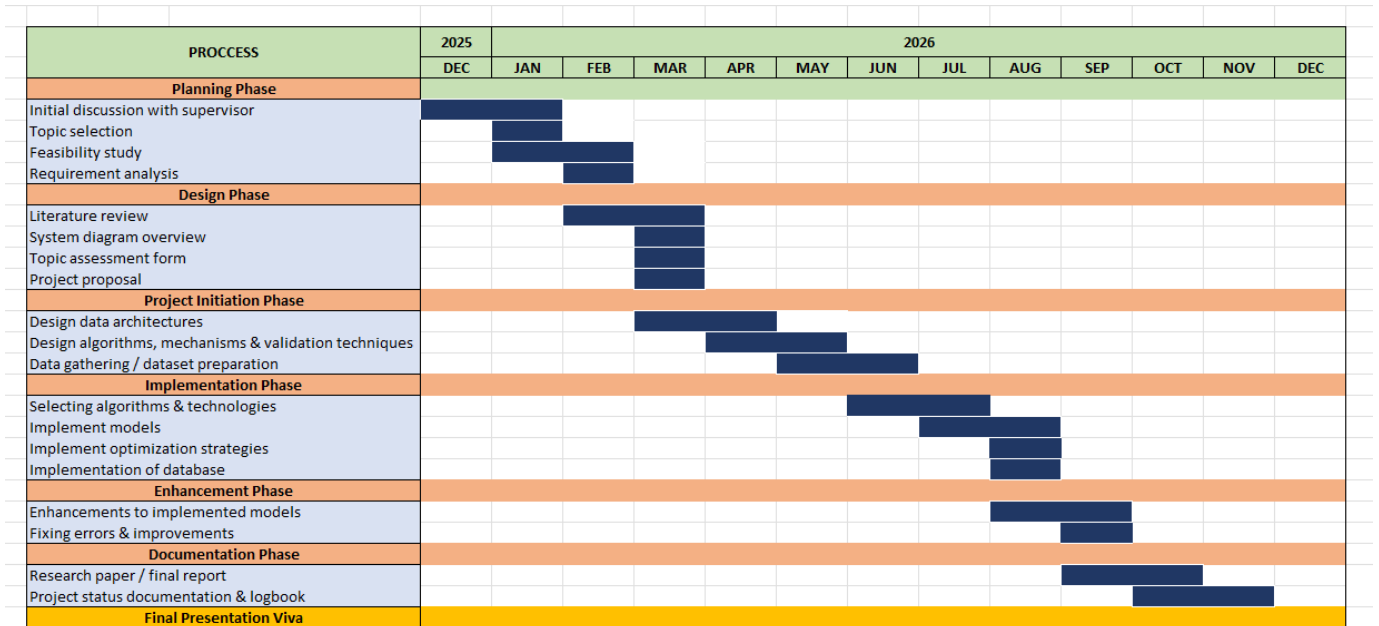


Fig.4 Gantt chart

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11. APPENDICES

Appendix A - Paddy Crop Growth Stages and Climate Sensitivity

TABLE VII Paddy Crop Growth Stages and Climate Sensitivity

Growth Stage	Approx. Duration	Key Farming Activity	Climate Influence
Land Preparation	0–15 days	Soil preparation	Rainfall
Vegetative Stage	15–45 days	Irrigation & fertilizer	Temperature
Reproductive Stage	45–75 days	Nutrient management	Rainfall
Maturity Stage	75–100 days	Harvest planning	Dry weather

Appendix B - Climate Variables Used in the Proposed System

TABLE VIII Climate Variables Used in the Proposed System

Variable	Description	Unit	Importance for Paddy Cultivation
Rainfall	Daily precipitation level	mm	Influences irrigation requirements
Temperature	Average daily temperature	°C	Affects crop growth rate
Humidity	Atmospheric moisture level	%	Impacts plant health
Solar Radiation	Sunlight exposure	W/m ²	Influences photosynthesis

Appendix C – Sown and harvested Extend - YALA

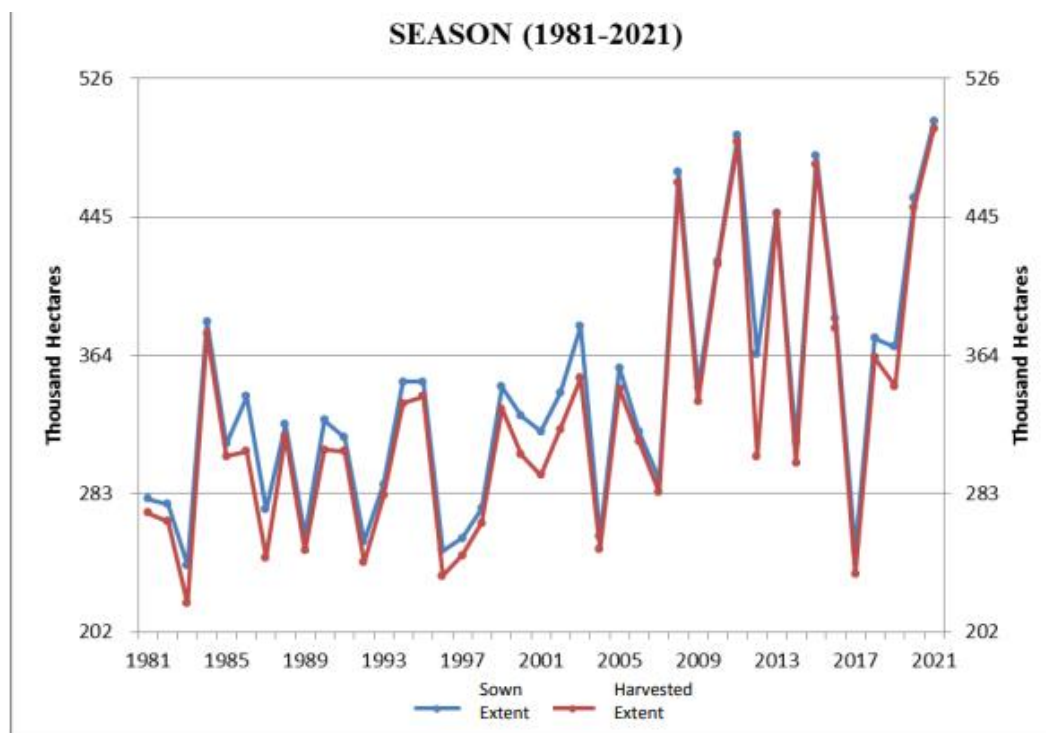


Fig.5 Sown and harvested Extend

Appendix D – Impact of climate change on paddy crop failure

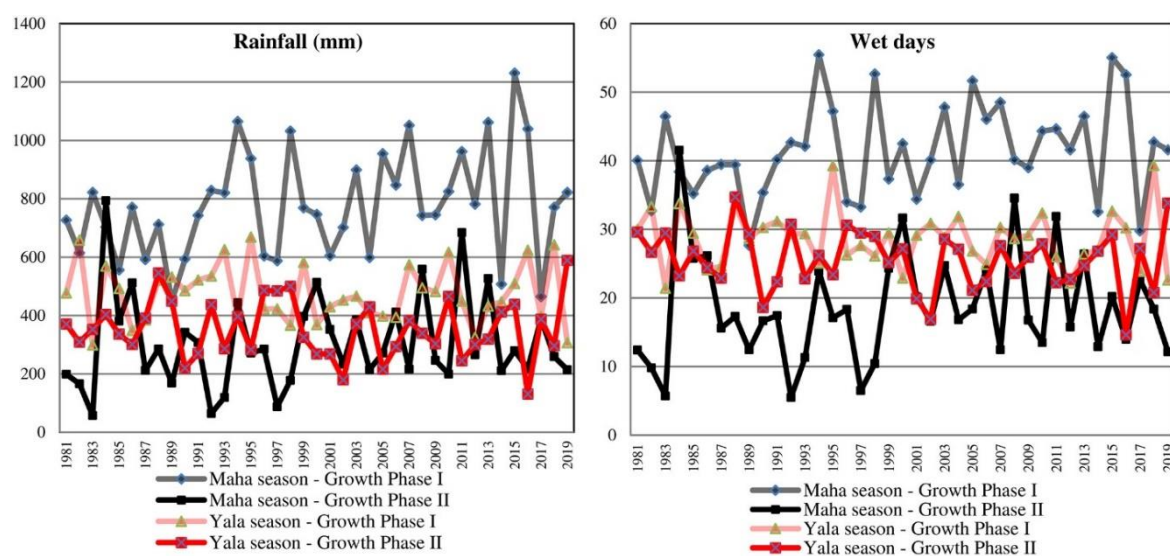


Fig.6 Impact of Climate Change on paddy Crop failure

AI-Driven Climate-Aware Decision Support and Risk Forecasting Systems for Sri Lanka

R26-IT-046

Project Proposal Report

Abhisheka K.K.R.H – IT22553232

B.Sc. (Hons) Degree in Information Technology Specialized in
Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka

March 2026

DECLARATION

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

Name	Student ID	Signature
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The above candidate is conducting research for their undergraduate Dissertation under my supervision.

Name	Date	Signature
Thisara Shyamalee	19 th March 2026	

ABSTRACT

Climate variability has become an increasingly important factor in transportation planning, especially in countries where road-based mobility supports daily commuting, logistics, emergency movement, and access to public services. In Sri Lanka, intense rainfall, urban waterlogging, flash floods, low visibility, and localized road blockages frequently interrupt travel and create unsafe road conditions. During such events, commuters often depend on a mixture of map applications, social media posts, weather updates, and informal local knowledge to make routing decisions. This fragmented decision environment increases uncertainty and can lead travelers toward routes that appear faster in traffic terms but are unsafe in practical terms. Although modern navigation systems provide travel-time optimization using live traffic conditions, most do not explicitly forecast short-term climate-induced road risk at road-segment level for Sri Lankan Road users.

This research proposes a Climate-Aware Safe Routing System as a component within an AI-driven climate-aware decision support platform for Sri Lanka. The proposed system integrates historical and near-real-time weather observations, forecast data, road network information, traffic-aware route conditions, and verified incident reports into a unified spatiotemporal dataset. Following data cleaning, normalization, spatial matching, and temporal alignment, the system applies a Long Short-Term Memory (LSTM) model to predict short-term disruption or hazard probability for individual road segments. These predictions are transformed into risk scores and incorporated into a graph-based routing framework using Dijkstra or A* so that route selection reflects both travel efficiency and predicted safety. Rather than simply recommending the fastest path, the system is designed to recommend the safest practically usable path under current and near-future conditions.

The expected outcome of this research is a pilot-ready prototype capable of improving commuter safety, supporting climate-resilient mobility, and assisting operational decision-making during adverse weather. The study also explores commercialization potential through a balanced model that keeps essential public-safety guidance accessible while creating revenue opportunities through enterprise dashboards, API services, and institutional partnerships. This makes the project not only technically relevant, but also socially and economically meaningful within the Sri Lankan context. Sri Lanka already has relevant public inputs such as DMC closed-road and situation reports, Department of Meteorology observations and forecasts, and the RDA national road network; Google's current coverage documentation also lists Sri Lanka for Traffic Layer, Driving Directions/Snap to Roads, Walking Directions, Two-Wheeled Vehicle Directions, and Real-Time Disruptions, which makes a pilot implementation technically credible.

Keywords— climate-aware routing, intelligent transportation systems, hazard prediction, LSTM, safe navigation, weather-aware mobility, Sri Lanka.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
ML	Machine Learning
LSTM	Long Short-Term Memory
ARIMA	Autoregressive Integrated Moving Average
RMSE	Root Mean Square Error
MAE	Mean Absolute Error
AUC	Area Under the Curve

1. INTRODUCTION

1.1. Background and Context

Transportation networks are highly sensitive to weather conditions because mobility decisions depend not only on road geometry and traffic volume, but also on visibility, road-surface quality, water accumulation, drainage performance, and the speed at which hazards emerge. In many developing countries, this sensitivity is even higher because road infrastructure, warning systems, and data integration mechanisms are still evolving. Sri Lanka presents exactly this kind of context. The country experiences seasonal and event-driven rainfall patterns that can quickly transform ordinary traffic delays into unsafe travel conditions. Flooded junctions, blocked culverts, temporary road closures, and landslide-prone access roads are practical realities rather than abstract risk scenarios. As a result, climate-aware mobility is becoming an increasingly important research and operational problem [1], [2], [3].

Road transportation remains central to everyday life in Sri Lanka. Daily commuting to work, schools, universities, hospitals, and markets depends heavily on road accessibility. Delivery services, taxis, buses, and emergency vehicles are equally affected by sudden weather-driven disruption. Because road transport is deeply embedded in both urban and interurban mobility, a disruption in even a small set of critical road segments can create wider ripple effects across travel time, safety, and service continuity. The Road Development Authority's national road network data and the Disaster Management Centre's regularly updated situation reports reflect how important road status and route continuity are to the functioning of the country. [2], [5].

The problem is not simply that bad weather exists. The deeper issue is that decision-making is fragmented. A driver may check a map application for traffic, another app for weather, and social media for road closures, yet still fail to obtain a single trustworthy answer to the most important question: Which route is safer right now, and which route is likely to become risky soon? This gap between available data and actionable route guidance is where the present research is positioned. The Department of Meteorology publishes 3-hourly observations, rainfall information, weather graphics, and forecasts, while the DMC publishes closed-road and situation reports. However, these streams are not automatically combined into a user-facing routing engine. [2], [3].

Existing navigation platforms are useful, but they are mostly optimized for speed and congestion reduction. Their underlying logic generally assumes that minimizing travel time produces the best route. In stable conditions, that assumption may be acceptable. In unstable weather conditions, however, the fastest route may pass through flooded low-lying roads, waterlogged urban junctions, or roads with rapidly worsening travel conditions [15], [16]. Therefore, what is needed is not just another navigation interface, but a decision-support mechanism that elevates safety into the routing objective itself.

This study responds to that need by proposing a climate-aware safe routing component that integrates weather data, traffic-aware travel information, crowd-verified incidents, and road-network intelligence into one predictive routing framework. The goal is to move from reactive routing toward proactive and risk-aware mobility support. In doing so, the study contributes not only to intelligent transportation research, but also to climate resilience, public safety, and digitally supported mobility planning within Sri Lanka [12], [13].

1.2.Literature Review

Research in intelligent transportation systems has long recognized that traffic conditions are dynamic and strongly influenced by temporal variation. Early prediction approaches often relied on statistical methods such as ARIMA, linear regression, or Kalman filtering. These methods were valuable for modeling short-term fluctuations when traffic patterns were relatively stable and data structures were simple [12]. However, they tended to struggle when nonlinear dependencies, abrupt changes, or multi-factor influences became dominant. This limitation became more visible as traffic systems began to be studied together with weather variation, accidents, event-driven congestion, and route-level disruptions.

The shift toward machine learning and deep learning emerged from this limitation. Recurrent models, particularly LSTM-based architectures, became attractive because they are better suited to sequence data and can capture temporal dependencies across multiple time steps. In traffic forecasting literature, LSTM models have shown promising results when historical speed, flow, occupancy, and time-of-day patterns are used together [12], [13]. Their main strength lies in their ability to retain useful temporal memory while avoiding some of the vanishing-gradient issues associated with simpler recurrent networks. As a result, LSTM models became a widely accepted option for short-term forecasting in transportation research.

A second important development in the literature is the growing recognition that weather must be treated as a first-class input rather than as an optional background variable. Rainfall affects traffic speed, braking distance, lane choice, driver caution, and network capacity. Poor visibility and wet road conditions can create congestion even when there is no physical road closure. Heavy rainfall can also turn already congested junctions into high-risk mobility points. Studies that incorporate weather features into prediction models generally report improved forecasting quality because the models are exposed to environmental drivers that explain sudden variation more realistically [13], [14]. This becomes especially important in tropical and monsoon-influenced regions, where rainfall intensity and location can shift rapidly.

Another branch of the literature goes beyond traffic forecasting and focuses on risk-aware or hazard-aware routing. These studies ask a different question from ordinary route optimization. Instead of minimizing only time or distance, they attempt to minimize exposure to danger, unreliability, or expected disruption. Flood-aware route-selection studies are particularly relevant because they demonstrate that the “best” route during extreme weather is not necessarily the shortest or fastest one [15]. Some studies show that a slightly longer route may reduce exposure to high-risk segments substantially, which is highly relevant for commuters, delivery operations, and emergency vehicles.

Graph-based routing research is also important for this proposal. Algorithms such as Dijkstra and A* are well established for shortest-path computation, but their practical value depends on how edge cost is defined. In ordinary map systems, the edge cost is often based on travel time or distance. In a climate-aware routing system, the edge cost must be reformulated to include predicted hazard risk, warning severity, or disruption likelihood [15], [16]. This transforms the routing problem from pure efficiency optimization into multi-criteria decision-making. That theoretical shift is essential for justifying the proposed system academically.

Commercial systems such as Google Maps, Waze, and traffic management platforms demonstrate the operational importance of live data integration. They show that users value route recommendations that adapt to changing conditions. However, most of these systems remain reactive. They respond to congestion, accidents, or user reports once a condition is already visible. They rarely provide a localized and predictive estimate [6], [8]. of near-future road risk driven by rainfall, flooding tendency, or official warning escalation. In addition, their global design does not automatically guarantee alignment with the specific climate and data realities of Sri Lanka.

Taken together, the literature suggests three important conclusions. First, predictive modeling is necessary because climate-related road conditions evolve over time. Second, weather-aware features improve the realism of transport forecasting. Third, routing must be redesigned when safety becomes a goal rather than a by-product. These conclusions strongly support the direction of the proposed research.

1.3. Research Gap

Although the literature provides strong foundations in traffic forecasting, environmental modeling, and route optimization, an important gap remains at the point where these areas intersect. Most existing studies concentrate on one piece of the problem rather than the full decision process [12], [13], [14]. Traffic forecasting studies often stop at predicting flow, speed, or congestion. Hazard studies often focus on exposure assessment or event analysis. Navigation systems often focus on ETA optimization. Few solutions combine multi-source climate-aware hazard forecasting with dynamic route generation in a way that is locally realistic for Sri Lanka.

The first gap is the reactive nature of existing navigation tools. Many map-based applications can reroute after congestion builds or after a user reports a blockage. However, a climate-aware safety system should ideally predict short-term risk before a commuter physically reaches the affected segment [6], [8]. For example, a route that looks acceptable now may become risky within the next 15–30 minutes due to rainfall buildup, drainage overload, repeated incident patterns, or a warning escalation. This future-oriented capability is still weak in most everyday navigation platforms.

The second gap is data fragmentation. Real-world road risk is not determined by one dataset. It emerges from the interaction between weather intensity, road type, localized traffic buildup, historical problem points, slope sensitivity, and incident occurrence. Sri Lanka has several useful information sources, including DMC reports, weather observations and forecasts, road-network data, and traffic-aware route services [2], [3], [5]. Yet these are not naturally aligned in one spatiotemporal pipeline. Building that integration layer is itself a research contribution because it requires schema design, geospatial matching, temporal synchronization, and label generation under imperfect data conditions.

The third gap is the lack of Sri Lanka-specific contextualization. Many published systems are developed for highly instrumented urban environments with dense sensors and clean streaming data. Sri Lanka does not always provide that level of instrumentation [1], [4]. Therefore, a locally relevant solution must work with a hybrid data strategy that combines official web portals, route APIs, historical road information, and verified reports. A model that assumes rich

sensors everywhere is less valuable than a model that performs credibly under realistic local data constraints.

The fourth gap is deployment realism and social inclusiveness. Even if a routing model is technically sound, it may fail as a public-facing solution if it depends too heavily on expensive commercial APIs or if essential safety guidance is hidden behind premium subscriptions [7], [10], [11]. A strong research contribution therefore needs to think beyond prediction accuracy alone. It must address affordability, explainability, accessibility, and phased deployment.

This study addresses these gaps by proposing a unified framework that combines data integration, short-term hazard forecasting, road-segment risk scoring, and dynamic routing for Sri Lanka. Its novelty lies not in using LSTM alone, nor in using Dijkstra alone, but in linking these elements within a climate-aware decision process that is operationally meaningful, technically feasible, and locally grounded.

2. OBJECTIVE

2.1 Main Objective

To design, develop, and evaluate an AI-driven Climate-Aware Safe Routing System for Sri Lanka that predicts short-term road hazards using weather, traffic, and incident-related data, and generates optimized travel routes that reduce both travel delay and exposure to unsafe road conditions.

2.2 Specific Objectives

- a) To identify and collect multi-source datasets relevant to climate-aware mobility in Sri Lanka, including weather observations, short-term forecasts, road network data, traffic-aware route information, and verified road incident reports.
- b) To preprocess these datasets by performing cleaning, normalization, spatial matching, temporal alignment, and feature extraction so that information from different sources can be represented consistently at road-segment level.
- c) To develop a predictive hazard forecasting model using LSTM networks that can learn temporal relationships among rainfall conditions, traffic anomalies, historical disruption patterns, and reported incidents.
- d) To transform predicted hazard probabilities into interpretable road-segment risk scores that can support route decision-making instead of being used only as abstract model outputs.
- e) To implement a graph-based climate-aware routing mechanism using Dijkstra or A* where route selection is based on a combined cost of travel efficiency and hazard exposure.
- f) To develop a mobile-oriented prototype interface that provides safe-route recommendations, near-real-time hazard alerts, and route updates during adverse weather conditions.

g) To evaluate the technical, operational, and commercial feasibility of the proposed system through prediction metrics, routing performance comparison, usability feedback, and budget analysis.

3. METHODOLOGY

3.1 Research Approach

This research adopts a Design Science Research (DSR) approach because the primary goal is to produce an artifact that solves a practical problem while also generating academic knowledge about design choices, system behaviour, and evaluation outcomes. DSR is suitable here because the study is not limited to describing climate-related transport problems; it aims to design and validate a working solution.

The research process will move through a series of connected stages. The first stage is problem framing, where the transport challenge, local context, stakeholder needs, and pilot region are defined. The second stage is data acquisition, where relevant climate, road, traffic, and incident-related information is collected. The third stage is data engineering, which includes preprocessing, spatial matching, temporal alignment, and feature generation. The fourth stage is model development, where the hazard prediction mechanism is designed and trained. The fifth stage is route optimization, where segment-level risk is incorporated into a graph-based routing engine. The sixth stage is prototype implementation, where a mobile-friendly interface is connected to the backend services. The final stage is evaluation, where the artifact is assessed from predictive, routing, usability, and feasibility perspectives.

To ensure realism, the implementation will begin as a pilot rather than a nationwide system. A pilot-oriented design is academically stronger because it avoids overclaiming scale while still allowing rigorous experimentation. It also makes it easier to explain data availability, evaluation boundaries, and budget assumptions in a credible manner.

3.2 Technical Methods and Algorithms

The technical core of the system combines spatiotemporal modeling and graph-based optimization.

At the data level, the road network will be represented as a graph in which intersections or junctions are nodes and road segments are edges. Each segment will carry both static and dynamic attributes. Static attributes may include road category, directionality, historical importance, and location-based characteristics. Dynamic attributes may include rainfall intensity, forecast severity, travel-time anomaly, recent incident presence, and predicted hazard probability.

The preprocessing stage will be especially important because the system depends on heterogeneous data. Weather data may arrive as station records, forecast summaries, or grid-based values. Traffic-aware data may arrive as route durations or segment-related speed variation. Incident data may be structured or semi-structured. Therefore, preprocessing will include missing-value handling, outlier checking, timestamp standardization, route-to-segment matching, and the creation of fixed time windows such as 15-minute or 30-minute intervals. These windows will serve as the temporal unit for feature generation and model learning.

Feature engineering will create variables that are meaningful for routing risk. Examples include current rainfall amount, rolling rainfall accumulation, short-term forecast intensity, change in travel time relative to a normal baseline, frequency of prior incidents on the same segment, warning severity, time of day, day of week, and local recurrence patterns. The purpose of feature engineering is not merely to increase model complexity, but to encode physically and operationally meaningful signals that relate to road hazard formation.

For prediction, the primary model will be an LSTM because the problem is sequential. A segment's hazard state at time t is influenced by recent rainfall, prior congestion buildup, earlier incidents, and short-term environmental trend. LSTM is well suited for capturing these dependencies. However, the study will not assume that LSTM is automatically the best model. A baseline comparison with simpler models such as Logistic Regression, Random Forest, XGBoost, or GRU-based variants will be included. This strengthens the methodological quality of the study because it shows that the final model choice is evidence-based rather than tool-driven.

The final route will be computed using either Dijkstra or A*. Dijkstra is reliable and well understood, while A* can improve efficiency when a suitable heuristic is available. The choice between them can be justified experimentally based on response time and route consistency. This part of the methodology is important because it links predictive analytics to a practical decision outcome, which is the route recommendation itself.

3.3 Tools and Platforms with Justification

Python is selected as the primary language because it supports the full research workflow, from ETL and geospatial preprocessing to model development and API integration. Libraries such as Pandas and NumPy are appropriate for data handling, while GeoPandas, Shapely, and OSMnx support spatial representation and road-network operations. NetworkX can be used for graph construction and route search logic.

TensorFlow/Keras is selected for LSTM development because it has strong support for sequence models, good documentation, and relatively smooth integration with deployment-oriented workflows. For this specific project, TensorFlow is preferable because the model pipeline is likely to move from experimentation into a lightweight inference service. PyTorch is also a valid alternative, but TensorFlow is selected because of its mature production-oriented ecosystem and accessibility for structured prototype development.

Firebase is a suitable backend option at prototype stage because it simplifies authentication, lightweight storage, real-time updates, and notification support. Importantly, Firebase currently offers a no-cost Spark plan and a pay-as-you-go Blaze plan, which makes it useful for a student-scale prototype before any large commercial scale-up.

React Native is selected for the mobile application because it enables faster cross-platform development compared to maintaining separate Android and iOS codebases. Since the primary aim of this research is to validate the routing concept rather than to engineer platform-specific UI complexity, React Native is a practical choice.

For maps and traffic-aware travel signals, Google Maps Platform can be used selectively. This is important: the proposal should present Google Maps Platform as a supporting input and visualization service, not as the intelligence of the system itself. Google's current

documentation shows pay-as-you-go pricing for core services and feature-based billing for Routes API usage, which is why API usage must be controlled carefully during research and any future commercialization.

For weather access, the system can rely primarily on official Sri Lankan weather sources where possible, with OpenWeather or similar services used as supplementary access points during prototyping. OpenWeather currently provides both free access and higher-tier plans, which supports a phased approach to development.

3.4 Data Collection and Ethical Considerations

The proposed data collection strategy is intentionally hybrid because no single source fully captures transport risk. The system will gather road-network data, weather observations and forecasts, traffic-aware route conditions, and incident information from multiple channels. The Department of Meteorology and DMC are especially important because they provide official public information relevant to weather conditions and disruptions. The RDA provides the national road network context, which is necessary for route modeling. These official sources improve local relevance and strengthen the credibility of the dataset design.

A practical challenge is that some of these sources are not naturally delivered as clean structured APIs. Therefore, the methodology must acknowledge the need for an intermediate ETL layer that extracts, standardizes, and stores relevant information in a model-ready form. Stating this limitation actually improves the proposal because it shows awareness of real implementation conditions rather than assuming perfect data availability.

Crowd-sourced incident reports can add valuable context, especially for sudden blockages or localized flooding. However, such reports create an ethical and operational issue: they may be noisy, delayed, duplicated, or inaccurate. For that reason, the system should not automatically trust every report equally. A simple validation mechanism should be described, such as confidence scoring, duplicate suppression, cross-checking with weather severity, or admin moderation for high-impact updates.

Privacy must also be treated seriously. The system does not need to retain detailed personal movement histories to function as a routing prototype. During experimentation, user-related data should be minimized, anonymized where possible, and retained only for the required evaluation period. Authentication, access-control rules, and restricted database access should be used to protect any operational data stored in the backend.

The proposal should also mention explainability ethics. If a system changes a user's route during bad weather, the user should receive a short explanation such as "route updated due to high rainfall and incident risk on the original segment." This matters because trust in a safety-support system depends not only on accuracy, but also on understandable communication.

3.5 Validation Strategy and Performance Metrics

The evaluation strategy should show that the system is not judged on model accuracy alone. A strong proposal demonstrates that success must be measured across prediction, routing, user experience, and feasibility.

At the prediction level, the hazard model can be evaluated using classification metrics such as Precision, Recall, F1-score, and ROC-AUC if the problem is formulated as hazard / non-hazard prediction. If the output is treated as a continuous risk score, MAE and RMSE can also be reported. A time-based train-validation-test split should be used because random shuffling is not appropriate for sequential prediction problems.

At the routing level, climate-aware routes should be compared with ordinary shortest-path or fastest-route recommendations. The comparison should focus on:

- average reduction in hazard exposure,
- percentage of risky segments avoided,
- additional travel time introduced for safer routing,
- route recalculation time,
- route stability under repeated updates.

This is important because a climate-aware system should not be evaluated only by whether it saves time. In some conditions, the safer route may reasonably take a little longer, and that trade-off must be measured explicitly.

At the usability level, the prototype should be evaluated through user-oriented testing. Test users should assess whether route alerts are understandable, whether rerouting explanations are convincing, and whether the application is easy to interpret under stressful or time-sensitive conditions. A short usability questionnaire and a few interview-based observations are enough for a student proposal.

Finally, at the feasibility level, the system should be assessed for operational practicality. This includes response latency, data freshness, dependence on paid services, and approximate per-climate-aware navigation recommendations to users through a mobile or web-based interface. Including this part in the proposal is a strong move because it connects the technical artifact to real-world sustainability.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview of the Proposed Solution

The proposed Climate-Aware Safe Routing System is designed as a modular decision-support architecture that collects, integrates, analyzes, and operationalizes multiple sources of mobility and environmental information. The architecture begins with data acquisition from weather, traffic, incident, and road-network sources. These inputs are processed into a unified spatiotemporal representation that supports both learning and inference. A prediction module estimates short-term hazard risk for road segments, and the routing engine uses that information to generate safe-route recommendations. The final recommendations are delivered to users through a mobile or web-based navigation interface.

The high-level architecture already shown in the diagram on page 19 is a good starting point because it clearly illustrates the flow from data sources to processing, hazard prediction, risk scoring, routing, and user delivery. However, to score better under the mark sheet, the text explanation needs to do more than label the blocks; it needs to justify why those blocks exist and how they interact operationally.

4.2. System Components

The architecture contains five major components.

1. Data Acquisition Layer

This layer collects data from official weather services, traffic-aware route services, incident-report sources, and road-network repositories. It may use scheduled pulls, API requests, lightweight web extraction, or manual feeds depending on source availability. The purpose of this layer is to ensure that the rest of the system receives timely and structured inputs.

2. Data Processing and Integration Layer

This layer performs cleaning, normalization, schema mapping, timestamp alignment, and geospatial matching. It is the most important integration layer because the data sources do not naturally share the same structure. By converting all inputs into road-segment and time-window representations, this layer enables both model training and real-time inference.

3. Hazard Prediction Module

This module consumes engineered features and predicts short-term hazard likelihood for each monitored road segment. The proposed implementation uses LSTM to capture temporal trends and interactions among weather and traffic variables.

4. Risk Scoring and Routing Engine

This layer transforms predicted hazard outputs into segment costs. A graph-search algorithm then computes the safest practical route according to the combined cost function. This is where the model's analytical output becomes a navigational decision.

5. User Interface Layer

This layer delivers safe-route recommendations, warning messages, route changes, and optionally incident reporting capability. It also communicates explanations that help users understand why the recommended route differs from the fastest route.

4.3. High-Level System Diagram

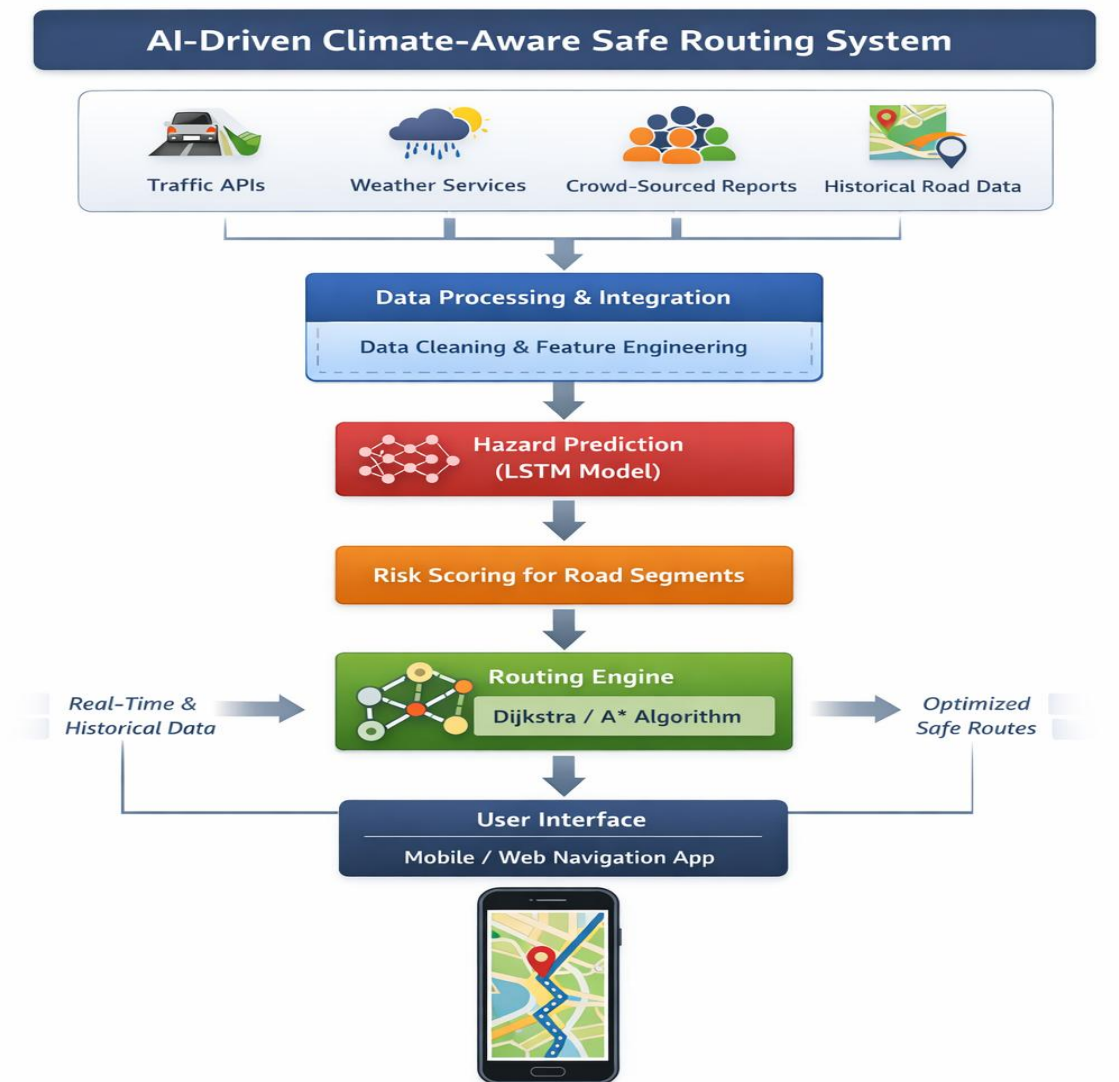


Figure 1 : High-Level System Architecture Diagram

4.4. Component Interactions

The interaction among components is continuous rather than one-directional. Data acquisition feeds new weather, route, and incident inputs into the processing layer. The processing layer updates the current segment-state representation. The hazard model then generates new risk estimates at each update cycle. These risk estimates are passed into the routing engine, which recomputes route cost when necessary. The user interface receives updated routes and alert messages. If user-confirmed incident reports are allowed, the feedback loop can send high-confidence updates back into the system for future route adjustments and dataset enrichment.

This cyclical interaction is important because safe routing is not a one-time calculation. Under bad weather, route conditions can change during the journey itself. Therefore, the system architecture must support repeated updates, not just static pre-trip planning.

4.5. Data Flow Explanation

The system follows a structured multi-stage data flow:

1. External services and repositories provide weather, road, traffic, and incident information.
2. Incoming records are validated and transformed into a consistent internal format.
3. Spatial and temporal alignment maps each record to relevant road segments and time windows.
4. Feature vectors are generated for model input.
5. The LSTM module predicts segment-level hazard probability.
6. Hazard probability is converted into routing cost.
7. The routing engine recalculates the safest practical route.
8. The user interface displays the route and related warnings.

This explanation should be written under the figure to show that the diagram is supported by a technically coherent process.

4.6. Scalability and Security Considerations

Scalability is addressed through modularity, caching, and phased deployment. Instead of monitoring every road in Sri Lanka from the beginning, the system can begin with selected high-risk corridors and later expand. Static road-network data can be preprocessed, while dynamic updates can be handled at interval-based refresh points. Cloud-based services can support burst workloads during heavy-weather events.

Security considerations include encrypted data transmission, backend access control, restricted API key exposure, authenticated reporting flows, and minimal retention of user-sensitive operational data. Because routing systems can indirectly reveal mobility patterns, privacy-conscious storage and log design are necessary even in a research prototype.

5. USER REQUIREMENTS

5.1. Stakeholder Identification

The system serves multiple stakeholder groups, each with different needs.

Daily commuters and private drivers need fast access to route recommendations that are easy to trust and understand. Their main pain point is uncertainty during rain, floods, or road closures.

Public transportation and ride-hailing operators need safe but operationally practical route updates. Their concerns include delay minimization, passenger safety, and service continuity.

Logistics and delivery companies need predictable mobility under disruption. For them, unsafe routing creates not only safety issues but also missed deliveries, fuel waste, and reputational damage.

Emergency services need route guidance that prioritizes access and safety under unstable conditions.

Government and disaster-management authorities may use the system as a situational awareness tool rather than as a consumer navigation app. For them, the value lies in identifying critical corridors, repeated disruption points, and operational risk zones.

Researchers and planners may use outputs for longer-term infrastructure and resilience analysis.

5.2. Functional Requirements

FR1: The system shall collect weather observations and forecast data relevant to road safety.

FR2: The system shall obtain traffic-aware travel-time information from external map or routing services.

FR3: The system shall collect and store incident reports from verified or moderated sources.

FR4: The system shall maintain a graph representation of the selected road network.

FR5: The system shall preprocess and integrate multi-source data into a unified spatiotemporal dataset.

FR6: The system shall generate engineered features for hazard prediction.

FR7: The system shall predict short-term road hazard probability using an ML model.

FR8: The system shall convert model predictions into road-segment risk scores.

FR9: The system shall compute climate-aware safe routes using graph-based optimization.

FR10: The system shall provide route updates when new data significantly changes risk conditions.

FR11: The system shall show hazard alerts and short explanations to end users.

FR12: The system shall support low-volume logging for evaluation and system improvement.

FR13: The system shall allow controlled administrative review of incident data if crowd reports are used.

FR14: The system shall support a pilot mode that limits the monitored geography and service cost.

5.2. Non-Functional Requirements

Performance: The system should generate route recommendations within an acceptable response time suitable for near-real-time navigation.

Reliability: The system should produce stable outputs under repeated queries and should not fail entirely if one optional data source becomes temporarily unavailable.

Scalability: The architecture should support gradual growth from pilot corridors to larger geographic regions.

Usability: Users should be able to understand route recommendations and hazard alerts with minimal training.

Availability: The service should remain accessible during adverse weather periods, which are precisely the times when it becomes most valuable.

Security: All communications between client, backend, and external services should use secure protocols, and operational data access should be controlled.

Maintainability: Data adapters and model services should be modular so that one service can be replaced without redesigning the whole application.

Explainability: The system should communicate short human-readable reasons for rerouting decisions.

Accessibility: The application should be designed for mobile-first use, clear visual communication, and future support for Sinhala and Tamil interfaces.

Affordability: Core safety functionality should be deliverable under a low-cost operational model, especially during the pilot stage.

6. COMMERCIALIZATION PLAN

6.1. Target Market and Customer Persona

The proposed Climate-Aware Safe Routing System has potential across both public-safety and commercial mobility markets. However, the most realistic commercialization strategy is not to depend only on individual consumer subscriptions. The stronger early market lies in stakeholders who incur measurable operational losses when routes fail.

The first customer persona is the daily commuter who wants reliable and understandable safe-route advice during heavy rain. This user values clarity and low cost.

The second persona is the ride-hailing or taxi driver who depends on route quality for daily earnings and is sensitive to both time loss and road risk.

The third persona is the fleet or logistics manager who needs route reliability at scale, with visibility into multiple moving vehicles and disruption-prone corridors.

The fourth persona is the public authority or disaster-management officer who benefits from a dashboard view of mobility risk rather than turn-by-turn navigation alone.

The initial market should focus on high-density or high-risk corridors where route disruption has visible operational consequences. This improves adoption likelihood and helps justify a pilot deployment.

6.2. Value Proposition

The system's central value proposition is that it provides safer route decisions under unstable weather, not just faster routes under ordinary traffic.

For commuters, the value lies in avoiding hazardous routes before they become visibly problematic. For drivers and operators, the value lies in reducing uncertainty, sudden delays, and risky exposure. For institutions, the value lies in transforming scattered road and weather information into decision-ready mobility intelligence.

What makes the value proposition stronger than a generic navigation product is the combination of:

- safety-oriented route choice,
- short-term predictive capability,
- local environmental awareness,
- operational usefulness for both public and enterprise users.

6.3. Revenue Model

A hybrid model is the most practical and socially acceptable strategy.

1. Free Public Safety Tier

Basic climate-aware safe-route suggestions, high-risk alerts, and limited rerouting support should remain free. This directly addresses the concern that low-income users may otherwise be excluded from essential safety information.

2. Premium Driver Tier

Frequent users such as ride-hailing drivers or daily commuters can be offered enhanced alert history, saved routes, commute monitoring, priority refresh, and offline-friendly features.

3. Enterprise / Fleet Tier

Logistics, delivery, and transport operators can be charged for dashboards, API integration, fleet visibility, route-risk analytics, and operational reporting.

4. Institutional Partnerships

Government agencies, local authorities, and NGOs may adopt the system for planning dashboards, sponsored public use, or event-based operational monitoring.

This model is stronger than a subscription-only approach because it cross-subsidizes public value through business and institutional usage.

6.4. Cost Estimation

The major operational costs involve cloud computing infrastructure, data storage, system maintenance, and technical support. Development Costs

Table 1 : Cost Estimation

Cost Component	Estimated Monthly Cost (LKR)	Annual Cost (LKR)
Cloud computing for model training and forecasting	6,000	72,000
Database and data storage services	3,000	36,000
Data backup and disaster recovery	2,500	30,000
Security and encryption services	1,500	18,000
System maintenance and monitoring	30,000	360,000

6.5. Pricing Strategy

The pricing model prioritizes institutional deployment while ensuring public accessibility to essential dengue warning services.

Table 2 : Pricing strategy

Service Type	Price (LKR)	Description
Government institutional license	1,800,000	Full system deployment, training, and integration
Enterprise organization license	1,400,000	Access to dengue risk analytics platform
Advanced climate–health analysis reports	180,000 per report	Custom outbreak risk assessment
Public web/mobile access	Free	Basic dengue alerts and hotspot maps

6.6. Competitive Advantage

The proposed system provides several competitive advantages compared to existing navigation platforms:

1. Climate-Aware Routing – Unlike conventional navigation systems that focus mainly on travel time optimization, this platform incorporates environmental risk factors such as rainfall and flooding.
2. Predictive Hazard Forecasting – The use of machine learning models enables the system to predict road hazards before they occur, allowing users to take proactive measures.
3. Localized Optimization for Sri Lanka – The system is specifically designed for Sri Lanka’s climate patterns, transportation infrastructure, and disaster risks.
4. Integration of Multi-Source Data – The system combines traffic data, meteorological forecasts, and crowd-sourced incident reports to provide a more comprehensive understanding of road conditions.

6.7. Estimated Project Budget

The development of the proposed AI-driven dengue risk forecasting system requires several technical and operational resources including computing hardware, cloud infrastructure, software tools, and human effort. These resources support activities such as data preprocessing, ML model training, system development, and visualization of predictive outputs. The pilot commercialization budget covers product hardening, limited field validation, cloud/API reserve, dashboard refinement, onboarding material, and contingency support for early deployment. This budget is not the same as the academic research budget. Instead, it reflects the initial resources required to move from a validated student prototype to a controlled external pilot.

6.8. Intellectual Property Considerations

Protecting intellectual property is essential for ensuring the commercial viability of the proposed system. The following measures may be considered:

1. Software Copyright Protection – The system’s source code, algorithms, and software architecture will be protected under software copyright laws.
2. Patent Opportunities – Novel aspects of the system, such as the integration of climate-based hazard prediction with routing algorithms, may qualify for patent protection if they demonstrate sufficient innovation.
3. Data Usage Agreements – Agreements will be established with data providers to ensure compliance with licensing conditions for traffic and meteorological data.
4. Trademark Registration – The platform name and branding may be registered as trademarks to establish a recognizable product identity.

By combining technological innovation with a well-defined commercialization strategy, the proposed system has the potential to evolve into a scalable smart mobility solution supporting climate-resilient transportation systems in Sri Lanka.

7. BUDGET AND JUSTIFICATION

7.1. Estimated Project Budget

This section presents the estimated budget required for the development and implementation of the Climate-Aware Safe Routing System. As this is a student-level research project, the budget is minimized by utilizing existing personal hardware, open-source development tools, and free-tier cloud services whenever possible. Most of the technologies used in this project, such as Python-based machine learning frameworks and geospatial processing libraries, are freely available, which significantly reduces development costs while still enabling the creation of a functional prototype.

7.2 Hardware Costs

The use of an existing laptop significantly reduces project costs.

7.3 Software Licenses

The system development primarily relies on free and open-source software tools widely used in data science and ML research.

Table 3 : Software Cost

Software	Purpose	License Type	Cost (LKR)
Python 3.x	Programming language for system development	Free / Open Source	0

TensorFlow / Keras	LSTM model development and training	Free / Open Source	0
GeoPandas & Network X	Geospatial processing and routing algorithms	Free / Open Source	0
VS Code	Code development and debugging	Free	0
Google Colab Pro	GPU-based machine learning model training	Paid (monthly)	25,000
TOTAL			25,000

Most of the development tools used in this research are open-source, which makes them ideal for academic projects. Google Colab Pro is included as a cost-effective alternative to purchasing dedicated GPU hardware, enabling efficient training of the LSTM prediction model.

7.4 Cloud Services

Cloud services may be required for data storage, model hosting, and visualization dashboards.

Table 4 : Cloud Services Cost

Service	Usage	Plan	Cost (LKR)
Firebase (Spark Plan)	Database and user authentication	Free tier	0
Firebase (Blaze Plan)	Cloud functions for model inference (if required)	Pay-as-you-go	5,000
Google Maps API	Map visualization and routing display	Free tier (limited)	0
Weather API (OpenWeather)	Weather forecast data for hazard prediction	Free tier	0
TOTAL			5000

Cloud services are necessary for data storage, map visualization, and real-time system interaction. The Firebase Spark Plan is sufficient for initial development and testing, while the Blaze Plan is reserved as a contingency for advanced backend processing if needed.

7.5 Human Effort Cost

These costs cover the printing and binding of the final research report and related documentation required for submission to the Department of Information Technology, SLIIT.

Table 5 : Human Effort Cost

Role	Task	Cost (LKR)
Printing & Binding	Research report and documentation submission	2,000
TOTAL		2,000

7.6 Justification of Expenses

Table 6 : Project Budget Summary

Category	Total Cost (LKR)
Software Licenses	25,000
Cloud Services	5,200
Human Effort Cost	2,000
GRAND TOTAL	LKR 32,000

The total estimated budget for this research project is LKR 32,000. This budget remains relatively low due to the use of open-source technologies, existing personal hardware, and free-tier cloud services. These cost-saving strategies allow the development of a functional AI-driven climate-aware routing prototype while keeping financial requirements manageable for a student-level research project.

7. WORK BREAKDOWN STRUCTURE (WBS)

The WBS is organized to reflect the natural progression of the research from problem framing to implementation and evaluation. Early tasks focus on proposal approval, literature review, scope confirmation, and data-source identification. Mid-stage tasks cover data collection, preprocessing, feature engineering, model development, and routing integration. Final-stage tasks include prototype validation, usability testing, report refinement, and viva preparation. This structure ensures that technical development is aligned with academic deliverables and that major dependencies, such as data readiness before model training, are respected.

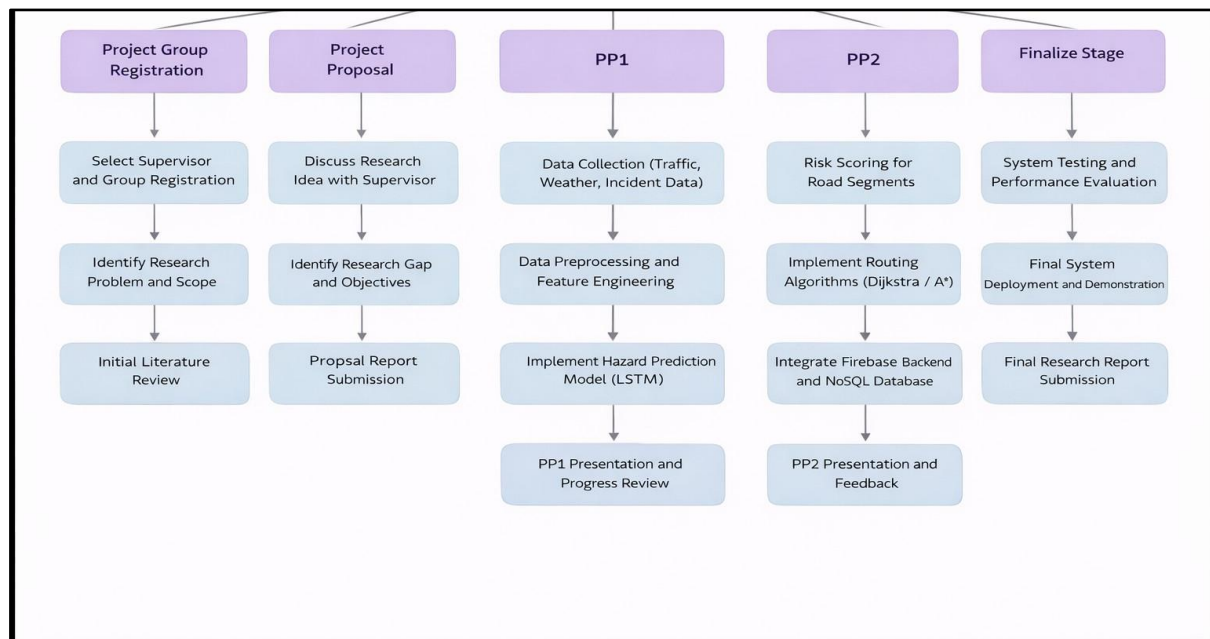


Figure 2 : Work Breakdown Structure

9. GANTT CHART

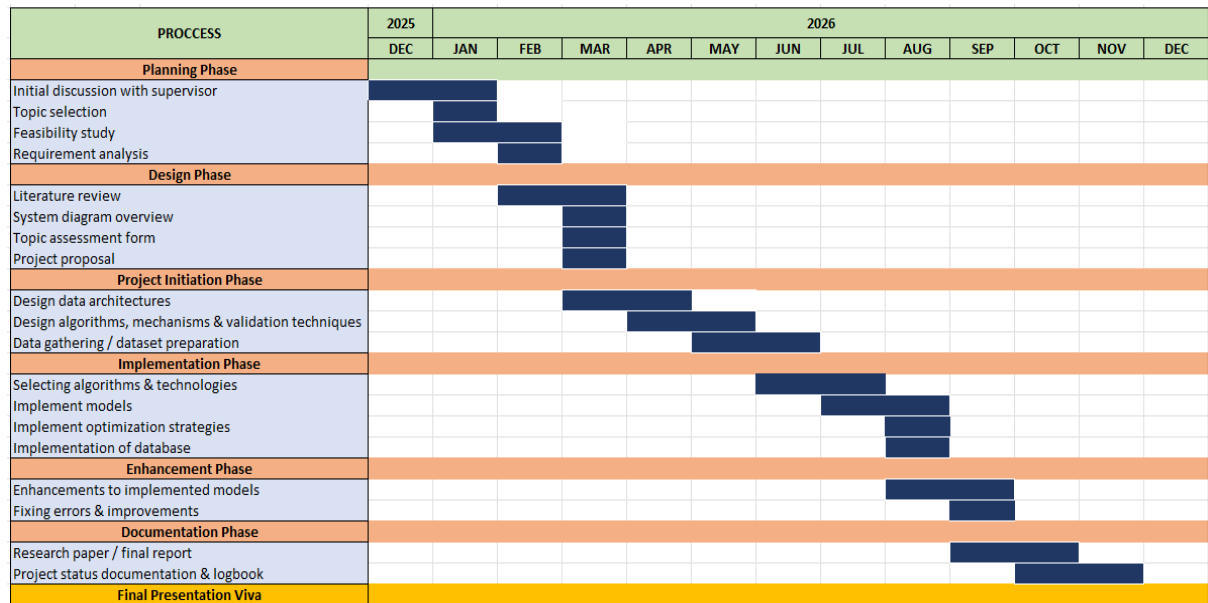


Figure 3 : Gantt Chart

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11. APENDICEIS

Appendix A - Sri Lanka main route map



Figure 4 : Sri Lanka Main Routes Map